AR-EAURP: Design, Native ns-3 Implementation, and Experimental Evaluation of Adversarial and Energy-Aware MANET Routing

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Abstract

Routing in mobile ad hoc networks must tolerate changing wireless links, limited energy, and insider forwarding attacks. AR-EAURP extends a trust- and energy-aware routing substrate with size-conditioned watchdog features, a clean-trained generative anomaly detector, harvested-power forecasting with a validated fallback, and a constrained deep-Q route selector. This paper describes the implemented native C++ system and evaluates its behavior in ns-3.45. Five routing-policy adaptations are compared through mobility, density, attack prevalence, energy, lifetime, component-ablation, constraint, and protection-overhead experiments. The review matrix completed 863 of 905 planned replications across 179 of 181 conditions; the density sweep remains partial. At 30% gray-hole prevalence, AR-EAURP achieved PDR 83.90% ± 8.08% compared with EAURP's 82.67% ± 2.84%; the paired difference was +1.23 percentage points, with a 95% interval of −9.06 to +11.53. Window-level gray-hole detection was weak (TPR 7.82% ± 4.98%, FPR 5.24% ± 5.14%, AUC 0.55 ± 0.07). One of eighteen unadjusted AR–EAURP attack contrasts excluded zero, while a separate thirty-pair commissioning study estimated +1.16 points (95% CI −2.97 to +5.30). Harvesting increased modeled residual energy, but the learned forecaster did not outperform persistence in the selected workload. The findings establish an auditable packet-level implementation and identify concrete detection and learning limitations; they do not establish general protocol superiority or field performance.

Keywords: mobile ad hoc networks; trust-aware routing; selective forwarding; generative anomaly detection; energy harvesting; constrained reinforcement learning; ns-3; reproducible evaluation.

1. Introduction

Mobile ad hoc networks (MANETs) exchange traffic through cooperating wireless relays without fixed infrastructure. Route quality depends on more than distance: a nominally short route may contain a depleted node, an unstable wireless link, or a relay that deliberately discards traffic. Trust-aware routing attempts to distinguish reliable forwarding from misbehavior, while energy-aware routing attempts to avoid exhausting the relays needed for future connectivity. Combining these objectives creates a difficult decision problem because the available measurements are local, delayed, and noisy.

EAURP motivates the combination of energy, trust, and protected communication [2]. The present project asks whether additional behavioral observations and learning components can improve that substrate under explicit packet-level attacks. Selective forwarding is particularly relevant: a relay can preserve a reasonable aggregate forwarding ratio by dropping only some packet sizes. An energy forecast may help identify a relay that will recharge, but a forecast that loses to persistence can add complexity without useful information. A constrained learner may react to delivery shortfalls, yet an unrealistic floor or poor feedback can also destabilize routing.

The project first studied these ideas in a Python simulation used for an earlier Colab review. That stage was useful for implementing components and producing exploratory comparisons. It could not establish performance in an IEEE 802.11 packet network. The current work therefore implements the system natively in ns-3, where route discovery, MAC/PHY delivery, queuing, overhearing, battery drain, and attacker decisions occur through simulated events. Moving to a richer model is an evaluation step; it does not guarantee that an earlier numerical advantage will persist.

The contribution is both an implemented routing system and a transparent empirical assessment. First, the paper specifies the actual native feature vector, model architectures, routing actions, training split, feedback, and shared-policy assumptions. Second, it compares five policy adaptations over explicit factor sweeps and reports completed run counts and missing conditions. Third, it separates anomaly detection from consensus and revocation and separates packet delivery from learner feedback. Fourth, it reports uncertainty, unsuccessful ablations, and historical results without treating them as proof of superiority. These details allow a reader to identify what is implemented, what is measured, and what remains unsupported.

The evaluation addresses four questions. RQ1 asks how delivery changes with mobility, density, attack type, and hostile fraction. RQ2 asks whether size-conditioned anomaly detection identifies selective forwarding and whether that translates into usable routing decisions. RQ3 asks which energy, forecasting, learning, and protection components justify their cost. RQ4 asks whether the observed AR–EAURP delivery differences are stable across matched replications. The answer is mixed: some point estimates favor AR, but most attack contrasts and the larger commissioning follow-up remain inconclusive.

2. Related work and positioning

AODV discovers routes on demand using route requests and replies [1]. It provides a useful conceptual reference for hop-based forwarding but does not by itself solve insider trust or energy management. The AODV-style policy evaluated here shares the project's custom routing harness. This controlled implementation comparison should be distinguished from benchmarking against stock ns-3 AODV, which is only used separately for clean integration checks.

Watchdog-based mechanisms infer forwarding behavior from overheard transmissions. Marti et al. [3] demonstrated the value of forwarding observation and path selection under routing misbehavior. In a wireless model, however, failure to overhear can have benign explanations. The current implementation binds evidence to the intended next hop and cancels pending evidence after a failed MAC handoff. Its contribution is the specific measured combination of evidence accounting, size-conditioned features, and corroboration, rather than a new claim that overhearing perfectly identifies malicious behavior.

Generative adversarial learning [4] and generative anomaly detection [5] motivate fitting a representation of normal behavior without training on the evaluation attack. AR-EAURP uses an adversarially trained generator and a nearest-generated-vector score. This is an implementation choice whose usefulness must be assessed against observed TPR, FPR, and AUC. Clean training alone is insufficient to establish robustness to mobility or selective attacks, and the detector is not a calibrated probability estimator.

LSTM models [6] can represent temporal dependence, while energy-harvesting studies emphasize the variability of replenishable sources [14]. Here the predictor is tested against persistence and a recent-window mean with a purged chronological split. This prevents the paper from attributing routing benefits to a neural predictor that was not selected. The harvesting process is a controlled synthetic source; it is not evidence about a specific outdoor energy system.

Deep Q-learning [7], Double Q-learning [8], and constrained Markov decision processes [9] motivate the route strategy agent. The implemented Lagrangian penalty responds to measured delivery shortfall but does not guarantee satisfaction of a long-run constraint under nonstationary topology. The experiments therefore include floor sensitivity, component removal, and matched run comparisons. The earlier Python variants, native DRL adaptations, and published EAURP are related artifacts with different provenance; none is silently treated as an exact external reproduction.

3. System model and implementation

3.1 Network and routing substrate

The system consists of mobile wireless nodes that originate application traffic, discover routes, and relay packets over an IEEE 802.11 ad hoc network. Each node maintains neighbor information, residual energy, routing state, and locally available forwarding observations. Packets traverse actual ns-3 IP, MAC, and PHY events. Delivery therefore depends on route discovery, queueing, retransmission, interference, mobility, relay behavior, and battery state. A packet is not declared delivered by a single network-wide probability. This distinction is central to the transition from the earlier Python study to the native experiment.

The implementation is a native C++ ns-3.45 module. The detector, forecaster, neural-network operations, tabular agents, and constrained route selector execute within that module. Python scripts aggregate saved outcomes and prepare reports; they do not replace the packet path with an external predictor. The protocol uses route requests, route replies, route errors, neighbor advertisements, trust reports, and revocation-related control messages. A route can fail before an attacker is reached, so malicious-drop counters and total application loss measure different processes.

For a neighbor j, forwarding trust is smoothed as $T_j^{new} = 0.7T_j^{old} + 0.3PFR_j$, where PFR is the observed forwarding ratio. The trust and energy route score has the form $R_j = \mathrm{clip}(a_i T_j + b_i E_j^{eff}, 0,1)$. The adaptive energy weight uses the mean normalized energy of the node and its nonexpired neighbor advertisements; the trust weight is its complement. The link cost includes $(1 - R_j)$ and a 0.35 hop penalty, represented internally in integer milli-cost units. This is a routing heuristic rather than a proof of an optimal path under the radio model.

Energy-aware variants exclude relays below 20% of their initial charge. The AODV-style reference uses equal hop costs and does not apply that energy gate. A contested AR neighbor receives an additional 0.5 trust multiplier for a 60 s contested interval. These choices can improve avoidance but also remove useful relays. Their possible cost is measured through PDR, delay, route failures, and component ablations.

3.2 Threat model

Attacks are insider forwarding and reputation attacks. Malicious nodes participate in the network and can forward control traffic while withholding relayed application packets. A black-hole relay drops relayed data. A gray-hole relay forwards smaller packets and independently drops eligible larger packets with probability 0.25, using a 700-byte size

threshold. The trust-poisoning configuration combines selective forwarding with dishonest reports: a colluder advertises favorable trust for fellow malicious nodes and low trust for honest nodes. The explicit collusion assumption allows an attacker to recognize its allies. Honest routing does not receive that ground-truth attacker list as an avoidance policy.

The experiment varies the configured hostile fraction from 0% to 50% in ten-percentage-point steps. With a finite number of nodes, configured prevalence is not a guarantee that every route encounters an attacker or that every adversarial node obtains a forwarding opportunity. Selective packet eligibility, sparse observations, and route changes further separate attack prevalence from the number of positive detector windows. Consequently, a 30% gray-hole condition is not equivalent to dropping 30% of all generated packets.

The evaluated scope excludes physical jamming, malicious radio hardware, cryptographic key theft, wormholes, and arbitrary compromise of the simulator itself. A low delivery ratio can still arise from benign link failure or a depleted relay. These competing explanations motivate careful watchdog accounting and an energy excuse before trust condemnation. No formal anonymity or traffic-unobservability property is established by the experiments.

3.3 Watchdog observations and detection layers

The watchdog associates a forwarded packet digest with the intended next hop and confirms an overheard onward transmission using the neighbor MAC binding. A failed link-layer handoff cancels its pending watchdog entry, avoiding the automatic attribution of that failed handoff to malicious forwarding. A missing observation is therefore handled differently from a confirmed successful forwarding event. The watchdog timeout is 1 s. Published opinions require sufficient accumulated evidence, and an anomaly window requires at least six new observations.

The native detector consumes nine features: short-window PFR, long-window PFR, their difference, large-packet drop ratio, small-packet drop ratio, scaled mean forwarding latency, scaled latency standard deviation, report disagreement, and the difference between large- and small-packet drop ratios. Latencies are scaled by 0.1 s and capped at two; features are sanitized and clipped to the implemented range. Size-conditioned observations target selective forwarding that can remain hidden in a single average PFR.

Three measurement layers are retained. First, window-level TPR, FPR, and AUC describe thresholded anomaly scores and score ranking on eligible windows. Second, contested-node measurements describe corroborated suspicion with honest-witness requirements. Third, revocation measurements describe the protocol's final exclusion state. A correct anomaly score is not automatically a corroborated accusation, and an accusation is not automatically successful rerouting. The consensus logic requires at least two accusers and at least half of the relevant observers, subject to local evidence and energy checks. Ground-truth labels support evaluation; they are not a substitute for this inference path.

3.4 Clean-trained generative anomaly model

The detector is trained during a separate 240 s attack-free commissioning simulation. For feature k, clean training statistics define $z_k = (x_k - \mu_k)/\max(\sigma_k, 10^{-3})$. At least 32 clean windows are required. The generator maps an eight-dimensional latent vector through two 32-unit ReLU layers to nine outputs. The discriminator uses layers of 32 and 16 units with leaky ReLU activations and a sigmoid output. Adversarial training uses 120 epochs, batches of 16, a learning rate of 2×10^{-4}, and one-sided real-label smoothing to 0.9. These are native implementation settings, not parameters copied from the older Python detector.

Following training, the generator produces a bank B of 2,048 standardized normal behavior vectors. The inference score is the nearest generated-vector root-mean-square distance:

$$s(x) = \min_{b \in B} \sqrt{\frac{1}{9}\sum_{k=1}^{9} (z_k - b_k)^2}.$$

The threshold is the 95th percentile of scores on the collected clean training windows. The score is not a maliciousness probability and is not the discriminator's real-versus-generated output. The model is related to generative anomaly modeling [4,5], but the generated-bank inference procedure is specific to this project. In particular, it does not optimize a latent vector separately for every evaluation sample.

Detector parameters are frozen for attack evaluation. This separates attack observations from model fitting, but the same clean collection is used for fitting and threshold calibration. The threshold therefore lacks an independent clean calibration set, and nominal 5% calibration does not guarantee a 5% FPR on new network conditions. Mobility, neighborhood changes, sparse evidence, and channel-induced losses can alter the score distribution. The measured evaluation FPR is reported instead of assuming the calibration target was achieved.

3.5 Harvest forecasting and fallback selection

The forecaster predicts mean harvested power over the next ten samples from a history of twenty samples. At each 1 s sample, the inputs are normalized harvested power and the sine and cosine of the modeled day phase. A 24-unit LSTM [6] is trained for 30 epochs with learning rate 0.005. Its target is power in watts, not cumulative battery energy. Routing blends current normalized energy and normalized predicted harvested power for its energy-oriented action.

For each node's commissioning trace, the earliest 70% supplies training data and the last 30% supplies validation. Sequences crossing the split are purged, and normalization uses training power only. Persistence and a recent-window mean are evaluated on the same held-out targets. The LSTM is used for routing only when its validation MAE is strictly below persistence; otherwise prediction falls back to persistence. The field historically named seasonal MAE corresponds to a window-mean predictor, so this paper does not call it a seasonal forecasting baseline.

All traces contribute to shared model parameters. There is one fleet-trained detector and one fleet-trained forecaster, rather than a separately trained network at each node. Provisioning such models is a plausible deployment step, but training communication, model distribution, and hardware inference cost are not evaluated as a distributed field system. Reporting the fallback outcome is necessary: an experiment in which persistence is selected cannot be cited as evidence that the LSTM improved delivery.

3.6 Constrained learning for route strategy selection

The CMDP-inspired agent chooses among trust-oriented, energy-surplus, balanced, and diversified routing strategies. Its six state variables are mean local trust, own residual-energy fraction, normalized speed, forecast surplus, contested-neighbor fraction, and rolling application feedback PDR. The feedback window contains the latest 100 delivery outcomes. Before feedback exists, its initial PDR value is one. This online state differs from the final measured application PDR and can be affected by timeout feedback.

The route selector uses a six-input, two-hidden-layer network with 64 ReLU units per layer and four action values. Its Double-DQN target [7,8] is

$$y = r + 0.9 Q_{target}(s', \operatorname{argmax}_a Q_{online}(s', a)).$$

The shaped reward is

$$r = (2I_{delivered} - 1) - 0.05 c_E - 5\lambda \max(0, \tau - \widehat{PDR}_{100}).$$

Here c_E is a hop-based energy proxy, and τ is the target delivery floor. The proxy derives from a 2 mJ-per-hop assumption and internal scaling; it is not the radio energy consumed by the packet. Actual battery behavior is measured separately through the ns-3 energy model. Every ten source attempts, the dual variable is updated as $\lambda \leftarrow \text{clip}(\lambda + 0.05(\tau - \widehat{PDR}_{100}), 0, 20)$.

The replay buffer holds 5,000 transitions; batch size is 32, learning rate is 0.001, and training is attempted every four observations. Target parameters use a hard copy every 200 training calls and a 0.01 Polyak update otherwise. Epsilon decays from 0.30 toward 0.05 with an 800-decision time constant. Exploration and shared policy updates continue during evaluation. The current implementation also learns from clean commissioning outcomes, rather than first activating the route learner under attack.

One CMDP agent is shared through the native experiment coordinator. This parameter sharing is a substantive simulation assumption: the study evaluates a coordinated fleet policy, not independently learning decentralized agents. The Lagrangian construction is motivated by constrained decision processes [9], but no convergence, constraint-feasibility, or safety guarantee follows from the implemented finite-run learner. The floor sweep explicitly measures whether higher targets translate into higher observed delivery.

3.7 Baseline definitions and protected payloads

All five comparison policies share the native packet harness. AODV denotes the custom equal-hop AODV-style reference; it is not ns-3's stock AODV implementation [1]. A separate ns3-aodv integration option exists for clean checks but is not included in the attack matrix. EAURP is the project's trust/energy substrate adaptation, inspired by the base protocol [2]. DRL-P and DRL-S denote native adaptations of the source-equation and delivered-implementation variants. Their tabular agent discretizes state, uses two actions, learning rate 0.1, discount 0.9, epsilon 0.1, and actual delivery feedback. Their behavior must not be described as verbatim reproduction of an external publication's complete original implementation.

Protected application traffic uses X25519 key agreement, HKDF-SHA256 key derivation, and AES-256-GCM authenticated encryption [10–12]. A 12-byte nonce and 16-byte tag add 28 bytes per protected payload. The measured aggregate overhead includes the packets counted by the native instrumentation; it is not simply the number of measured

source packets multiplied by 28. Key provisioning is trusted in the simulation. Payload protection does not prove authentication of every routing advertisement or traffic-analysis resistance. The no-ECC label in saved outputs disables the implemented payload protection path and should be interpreted in that operational sense.

4. Experimental methodology

4.1 Radio, workload, mobility, and energy settings

The experiments use ns-3.45 [13] with IEEE 802.11b ad hoc operation and fixed 2 Mbit/s DSSS transmission. Log-distance path loss has exponent 3 and reference loss 40.05 dB at 1 m, followed by Nakagami fading. Transmit power is 16.0206 dBm; receiver sensitivity is −96 dBm. The 150 m distance used in connectivity diagnostics is not a hard reception cutoff. This is a simulated wireless environment with channel and packet effects, rather than physical deployment evidence.

Table 1 records the base configuration. Random waypoint motion uses a uniform speed from 1 m/s to the selected maximum and pauses up to 2 s. Packet sizes are uniformly drawn from 512–1,024 bytes, so the gray-hole size rule sees both eligible and ineligible packets. Six CBR flows generate one packet per second each. The base evaluation horizon is 120 s, of which 20 s is warm-up. The actual metric denominator is 100 s, corresponding to 600 measured source packets. An additional 10 s cooldown permits final deliveries. AR clean commissioning is separate from this horizon. The 900 s lifetime condition has 880 s after warm-up.

Table 1. Native base settings and evaluation boundaries.

Setting	Value
Engine and implementation	ns-3.45; native C++
Base network	24 nodes; six CBR flows; 420 × 420 m
Traffic	1 packet/s/flow; uniform 512–1,024 bytes
Evaluation timing	120 s scenario; 20 s excluded warm-up; 100 s measured; 10 s cooldown
Commissioning	Separate clean 240 s simulation for AR models and route learning
Mobility	Random waypoint; 1–20 m/s base range; 0–2 s pause
Radio	802.11b ad hoc; 2 Mbit/s DSSS; 16.0206 dBm
Propagation	Log-distance exponent 3; Nakagami fading; constant-speed delay
Initial charge	350 J per node
Harvesting	70% panel-equipped nodes; 600 s modeled day; solar and kinetic input
Review RNG	Seed 20260909; matched RngRun 1–5
Default AR floor	0.90; continuing online learning and exploration
Batch pacing	Simulated time; wall-clock pacing disabled

The battery uses the ns-3 Wi-Fi radio energy model. Harvesting combines a modeled solar input and a speed-dependent kinetic term. The solar process uses a 1.2 W panel peak, a compressed 600 s day, and temporally correlated cloud variation; the kinetic coefficient is 0.004 W per m/s. These assumptions allow controlled comparisons within the model but do not calibrate a particular panel, climate, battery chemistry, or radio device [14]. Residual-energy results are network totals in joules, not energy per node.

4.2 Factor sweeps and actual completion

The design varies mobility (E1), density (E2), attack type and prevalence (E3), harvesting (E4), lifetime (E4b), component removal (E5), delivery-floor selection (E5b), and payload-protection overhead (E6). These are structured factor sweeps, not an exhaustive Cartesian product of every parameter. E1, E2, E4, and E4b use their clean scenario settings; E5, E5b, and E6 use 30% gray-hole evaluation. Exact overrides are preserved in the condition and run ledgers.

Table 2. Completed experimental coverage; missing runs are not imputed.

Experiment	Factor levels	Conditions completed/planned	Runs completed/planned
E1 mobility	Five policies × 1, 5, 10, 15, 20 m/s	25/25	125/125
E2 density	Five policies × 60, 80, 100, 120, 150, 200 nodes	28/30	108/150
E3 attacks	Five policies × three attacks × 0–50%, step 10%	90/90	450/450
E4 energy	Five policies × battery/harvesting	10/10	50/50
E4b lifetime	Five policies × energy mode; 900 s horizon	10/10	50/50
E5 ablation	Full and five component-removal settings	6/6	30/30
E5b delivery floor	0.70, 0.80, 0.90, 0.95	4/4	20/20
E6 overhead	Five policies and AR without payload protection	6/6	30/30
Total	Structured review matrix	179/181	863/905

The incomplete portion is confined to E2. All five policies have five runs at 60, 80, 100, and 120 nodes. All have one run at 150 nodes. At 200 nodes, only AODV, DRL-P, and DRL-S have one run; EAURP and AR are absent. The 42 deferred replications were stopped by a time limit. Their absence precludes a complete high-density comparison and may bias a summary that ignores the coverage pattern. A single-run outcome is shown as descriptive, without an empirical standard deviation or confidence interval.

4.3 Endpoints and statistical analysis

PDR equals the number of uniquely received measured application packets divided by measured generated packets. Delivered packets may arrive during cooldown. Throughput is received application bytes times eight divided by measured seconds. Mean delay is computed over delivered packets, so it is conditional on survival: an attacker can lower measured delay by eliminating difficult packets. A lower delay paired with much lower PDR is not automatically a routing improvement. Application delivery and the route learner's timeout feedback are distinct counters and can disagree for late deliveries.

Window-level TPR is TP/(TP+FN), FPR is FP/(FP+TN), and AUC summarizes ranking of anomaly scores when both classes are available. Clean-only conditions lack positive attack windows; a stored zero TPR or default AUC of 0.5 is not an observed successful detector. Baselines without the AR detector have no comparable detector endpoint. Repeated windows from the same node are dependent; the independent unit for reported variability is the simulation replication, not the number of packets or windows.

Each complete condition has five stochastic replications using one fixed master seed and distinct ns-3 run numbers. These are five run streams, not five distinct master seeds. Means and sample SDs summarize run-level measurements. Two-sided 95% Student t intervals use $\bar{x} \pm t_{0.975,n-1} s/\sqrt{n}$. Rate intervals in the saved tables are clipped to the allowed range; this is descriptive bounding, not a binomial interval construction. With five runs, estimates remain sensitive to topology and outliers.

For a matched protocol comparison, $d_r = 100(PDR_{AR,r} - PDR_{EAURP,r})$ is measured in percentage points. The same t construction is applied to these paired differences. Matching seed and run number reduces configuration mismatch but does not enforce identical event-level random draws: protocol execution changes random-stream consumption. E3 contains 18 AR–EAURP rows and 180 all-policy contrasts. Intervals are unadjusted, exploratory, and not a family-wise claim of significance. The zero-hostile scenario is repeated under three attack labels and must not be counted as three independent clean validations.

Lifetime endpoints are time to first death and time until 80% of nodes have died. If an endpoint has not occurred by the horizon, the observation is right-censored. Censored horizon or sentinel values are not treated as observed event times. The present report describes event flags and observed endpoints, without claiming a full survival analysis. The separate 30-pair commissioning follow-up uses seed 1 and run numbers 21–50 and is kept separate from the review matrix. Its larger sample size does not increase the number of independent replications in any E3 cell.

5. Results

Unless stated otherwise, table entries are run means ± sample SD and PDR/TPR/FPR values are percentages. DRL-P and DRL-S correspond to the saved labels DRL-EAURP (paper) and DRL-EAURP (senior). All comparisons concern the defined native adaptations.

5.1 Mobility and density

At 1, 5, 10, 15, and 20 m/s maximum speed, AR PDR was 86.60 ± 14.32%, 94.10 ± 5.56%, 83.93 ± 11.10%, 86.17 ± 6.90%, and 89.23 ± 4.73%, respectively. EAURP produced 86.80 ± 9.64%, 95.00 ± 4.99%, 88.10 ± 7.01%, 88.90 ± 6.38%, and 87.07 ± 6.41%. Neither sequence is monotonic. The results do not support a uniform mobility advantage for AR; at several speeds the simpler references have higher descriptive PDR. Figure 1 shows the five-policy comparison.

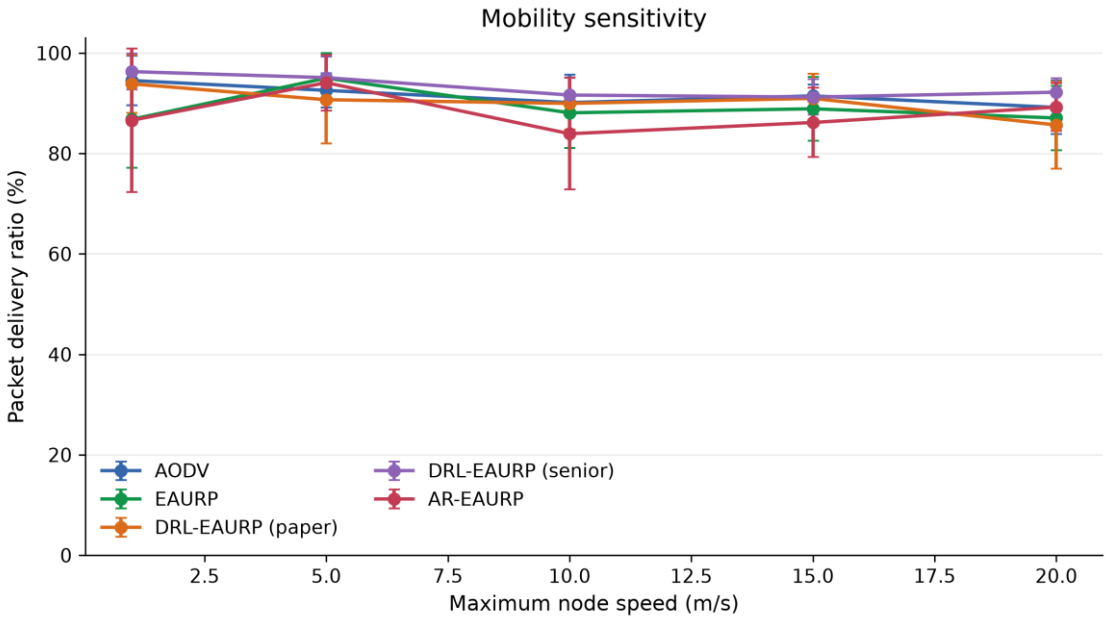


Figure 1. Mobility sweep. Points are run means; error bars show ± one sample SD. Each condition has five replications.

At 60 and 80 nodes, AR's PDR means were 92.40% and 88.00%, compared with EAURP's 89.47% and 87.03%. At 100 and 120 nodes, AR fell to 67.87% and 60.70%, compared with 70.00% and 65.50%. At 120 nodes the between-run SDs were about 19 points for both protocols. These large variations limit fine rankings. The 150-node AR observation was 34.17% with n=1; no AR result exists at 200 nodes. Figure 2 retains this incomplete coverage.

The density sweep changes node count at fixed area and six-flow load. Increased neighbor interactions, discovery activity, channel contention, and route-policy responses are plausible contributors to degradation, but this factor sweep does not isolate their causal effects. It also changes traffic per node. It would be incorrect to present the partial high-density points as a replicated scalability limit or to infer a universal node-count threshold.

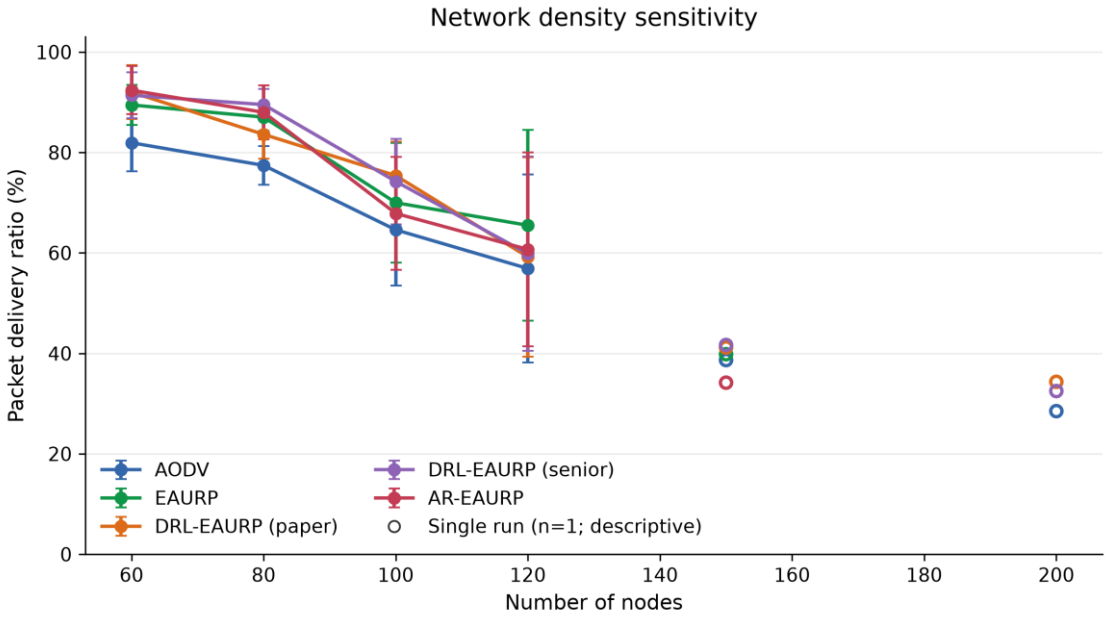


Figure 2. Density sweep with incomplete high-density replication. The settings ledger supplies actual n; single-run points are descriptive.

5.2 Delivery under black-hole, gray-hole, and trust-poisoning attacks

Tables 3–5 provide every hostile-percentage PDR result. Black holes cause a pronounced delivery collapse for all policies. At 30% hostile nodes, AR delivered 38.47% compared with EAURP's 40.80%; at 50%, both were near 21–22%. AR's high black-hole window TPR therefore does not imply preserved connectivity. Correct local detection can occur after losses, without enough trustworthy alternative relays, or without adequate corroboration. The measured result is poor delivery despite strong separation of the most obvious anomaly class.

Table 3. Black-hole PDR (%) across all five native policies; n=5 per cell.

Hostile nodes	AODV	EAURP	DRL-EAURP (paper)	DRL-EAURP (senior)	AR-EAURP
0%	89.17 ± 5.34	87.07 ± 6.41	85.67 ± 8.71	92.20 ± 2.78	89.23 ± 4.73
10%	44.80 ± 3.62	71.33 ± 15.58	71.67 ± 13.16	41.27 ± 10.03	63.67 ± 16.29
20%	28.57 ± 9.14	45.80 ± 23.59	50.17 ± 14.74	34.93 ± 9.25	45.47 ± 18.08
30%	30.37 ± 6.55	40.80 ± 15.61	38.40 ± 14.38	25.97 ± 9.38	38.47 ± 15.13
40%	23.13 ± 8.54	30.33 ± 15.43	29.30 ± 15.53	24.20 ± 12.06	33.77 ± 10.16
50%	24.80 ± 11.27	20.53 ± 9.06	25.07 ± 9.73	21.77 ± 11.73	21.83 ± 7.74

Gray-hole delivery degrades less sharply because only eligible larger packets are selectively dropped. At 30%, AR's mean PDR was 83.90% versus EAURP's 82.67% and DRL-S's 86.40%. AR is not the best descriptive policy in that cell. Its paired advantage over EAURP was +1.23 points with a wide interval spanning appreciable benefit and harm. At 20%, AR's point advantage was +6.67 points, but the interval still included zero. At 40% and 50%, the AR–EAURP means were negative. This pattern does not support a consistent prevalence-dependent improvement.

Table 4. Gray-hole PDR (%) across all five native policies; n=5 per cell.

Hostile nodes	AODV	EAURP	DRL-EAURP (paper)	DRL-EAURP (senior)	AR-EAURP
0%	89.17 ± 5.34	87.07 ± 6.41	85.67 ± 8.71	92.20 ± 2.78	89.23 ± 4.73
10%	87.10 ± 7.19	90.53 ± 5.05	89.50 ± 4.62	90.83 ± 4.44	87.43 ± 7.86
20%	84.27 ± 5.43	80.70 ± 8.88	84.63 ± 5.76	89.30 ± 5.79	87.37 ± 6.72
30%	83.67 ± 9.62	82.67 ± 2.84	82.60 ± 10.65	86.40 ± 6.85	83.90 ± 8.08
40%	82.50 ± 7.31	83.33 ± 8.92	83.03 ± 5.88	85.23 ± 7.72	81.23 ± 4.75
50%	81.13 ± 7.15	83.63 ± 9.21	75.10 ± 8.81	85.10 ± 5.73	80.17 ± 8.99

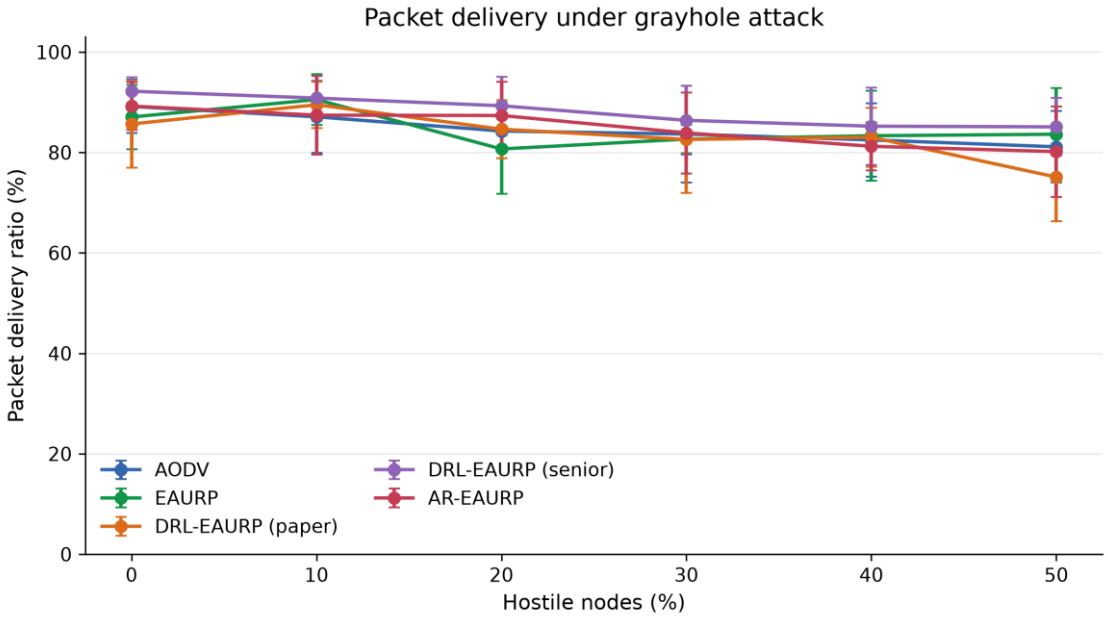


Figure 3. Gray-hole prevalence sweep. A positive point difference at one percentage is not a general performance guarantee.

Trust poisoning has a larger impact on the trust-sensitive policies than on the equal-hop reference. At 30%, EAURP and AR obtained 56.90% and 54.60%, while the AODV-style reference and DRL-S retained the same aggregate PDR as their gray-hole configurations. This is consistent with differences in how policies use the poisoned reports; it is not proof that those policies resist every reputation attack. At 20%, AR exceeded EAURP by +11.07 points, with an unadjusted 95% paired interval of +1.35 to +20.79. This is the only one of the eighteen displayed AR–EAURP intervals excluding zero. Its exploratory status and the lack of multiplicity adjustment prevent a broad superiority claim.

Table 5. Trust-poisoning PDR (%) across all five native policies; n=5 per cell.

Hostile nodes	AODV	EAURP	DRL-EAURP (paper)	DRL-EAURP (senior)	AR-EAURP
0%	89.17 ± 5.34	87.07 ± 6.41	85.67 ± 8.71	92.20 ± 2.78	89.23 ± 4.73
10%	87.10 ± 7.19	83.53 ± 9.72	78.83 ± 11.29	90.83 ± 4.44	80.47 ± 8.18
20%	84.27 ± 5.43	61.13 ± 14.01	78.47 ± 11.74	89.30 ± 5.79	72.20 ± 17.50
30%	83.67 ± 9.62	56.90 ± 14.72	70.47 ± 10.84	86.40 ± 6.85	54.60 ± 12.97
40%	82.50 ± 7.31	48.13 ± 12.71	57.83 ± 8.58	85.23 ± 7.72	58.00 ± 14.81
50%	81.13 ± 7.15	48.50 ± 11.02	54.00 ± 7.28	85.10 ± 5.73	47.03 ± 14.20

5.3 Detection, corroboration, and paired uncertainty

Table 6 combines the anomaly measurements with all eighteen matched AR–EAURP delivery differences. Black-hole detection has mean window TPR of roughly 93–100% at nonzero prevalence, while gray-hole TPR remains about 2–10%. At 30% gray holes, AUC is 0.55 ± 0.07 and FPR is 5.24% ± 5.14%, indicating weak score separation in this workload. Trust-poisoning AUC also remains close to chance and falls below 0.5 in some cells. These observations do not justify describing the native detector as highly accurate against selective or reputational attacks.

Table 6. AR window-level detector metrics and paired PDR differences; percentage endpoints except AUC. Intervals are unadjusted and exploratory.

Attack	Hostile nodes	AR window TPR	AR window FPR	AR score AUC	AR−EAURP PDR (pp)	Paired 95% CI (pp)	AR wins
blackhole	0%	0.00 ± 0.00	5.51 ± 2.94	0.50 ± 0.00	2.17	[-4.68, 9.02]	4/5
blackhole	10%	96.00 ± 8.94	6.63 ± 3.82	0.98 ± 0.02	-7.67	[-21.75, 6.41]	2/5
blackhole	20%	100.00 ± 0.00	7.26 ± 6.98	0.97 ± 0.03	-0.33	[-9.36, 8.69]	2/5

Attack	Hostile nodes	AR window TPR	AR window FPR	AR score AUC	AR−EAURP PDR (pp)	Paired 95% CI (pp)	AR wins
blackhole	30%	93.42 ± 8.17	10.97 ± 5.12	0.97 ± 0.04	-2.33	[-11.33, 6.66]	2/5
blackhole	40%	100.00 ± 0.00	13.69 ± 13.12	0.97 ± 0.06	3.43	[-9.16, 16.03]	2/5
blackhole	50%	99.05 ± 2.13	12.29 ± 12.89	0.98 ± 0.05	1.30	[-6.35, 8.95]	2/5
grayhole	0%	0.00 ± 0.00	5.51 ± 2.94	0.50 ± 0.00	2.17	[-4.68, 9.02]	4/5
grayhole	10%	6.19 ± 8.52	5.89 ± 3.45	0.55 ± 0.15	-3.10	[-8.64, 2.44]	1/5
grayhole	20%	2.36 ± 3.24	3.26 ± 2.49	0.62 ± 0.11	6.67	[-3.90, 17.24]	4/5
grayhole	30%	7.82 ± 4.98	5.24 ± 5.14	0.55 ± 0.07	1.23	[-9.06, 11.53]	2/5
grayhole	40%	9.83 ± 5.31	4.04 ± 4.53	0.59 ± 0.12	-2.10	[-12.75, 8.55]	2/5
grayhole	50%	9.29 ± 5.79	3.94 ± 3.54	0.56 ± 0.11	-3.47	[-10.35, 3.42]	1/5
trust poisoning	0%	0.00 ± 0.00	5.51 ± 2.94	0.50 ± 0.00	2.17	[-4.68, 9.02]	4/5
trust poisoning	10%	3.33 ± 7.45	7.40 ± 1.32	0.46 ± 0.17	-3.07	[-16.22, 10.09]	3/5
trust poisoning	20%	8.30 ± 6.04	7.64 ± 5.16	0.51 ± 0.03	11.07	[1.35, 20.79]	5/5
trust poisoning	30%	12.31 ± 8.77	5.00 ± 4.06	0.51 ± 0.13	-2.30	[-14.62, 10.02]	2/5
trust poisoning	40%	10.94 ± 3.96	7.70 ± 2.95	0.49 ± 0.04	9.87	[-1.37, 21.10]	4/5
trust poisoning	50%	10.17 ± 3.20	7.21 ± 5.36	0.39 ± 0.05	-1.47	[-21.89, 18.95]	2/5

For the 30% gray-hole condition, the separately recorded contested TPR was 0%, contested FPR 1.18%, revocation TPR 14.29%, and revocation FPR 11.76% as run means. These metrics use different event/state definitions and denominators from anomaly windows. Their disagreement is informative: a window score, a corroborated accusation, and a final trust exclusion describe different stages of the routing mechanism. In particular, a revocation can arise from ordinary trust logic without a successful GAN accusation. Reporting only the most favorable layer would conceal this distinction.

The three clean rows share the same underlying configuration and default no-positive-class behavior. Their TPR/AUC values cannot validate sensitivity. Across nonzero attacks, there is no fixed ordering of PDR that follows from detector TPR alone. The results support further work on observable evidence and robust scoring, rather than a claim that adding a neural detector automatically improves routing.

5.4 Energy and lifetime

Table 7 shows clean energy-mode results. AR residual energy increased from 6,022.38 ± 2.18 J without harvesting to 7,639.45 ± 207.81 J with harvesting. PDR changed from 87.47 ± 4.41% to 89.23 ± 4.73%. Similar residual-energy increases occur for all policies. Because added environmental energy directly changes the battery balance, the energy increase should not be attributed entirely to better route selection. No nodes died during the short E4 runs.

Table 7. Energy-mode PDR, total network residual energy, and dead-node count.

Configuration	PDR	Residual energy (J)	Dead nodes
AODV / battery	89.17 ± 5.34	6022.93 ± 2.25	0.00 ± 0.00
AODV / solar	89.17 ± 5.34	7640.87 ± 207.91	0.00 ± 0.00
EAURP / battery	88.83 ± 2.98	6021.79 ± 1.85	0.00 ± 0.00
EAURP / solar	87.07 ± 6.41	7640.65 ± 207.70	0.00 ± 0.00
DRL-EAURP (paper) / battery	82.93 ± 13.70	6021.05 ± 4.12	0.00 ± 0.00

Configuration	PDR	Residual energy (J)	Dead nodes
DRL-EAURP (paper) / solar	85.67 ± 8.71	7638.78 ± 208.57	0.00 ± 0.00
DRL-EAURP (senior) / battery	89.10 ± 4.57	6021.75 ± 2.39	0.00 ± 0.00
DRL-EAURP (senior) / solar	92.20 ± 2.78	7639.39 ± 207.71	0.00 ± 0.00
AR-EAURP / battery	87.47 ± 4.41	6022.38 ± 2.18	0.00 ± 0.00
AR-EAURP / solar	89.23 ± 4.73	7639.45 ± 207.81	0.00 ± 0.00

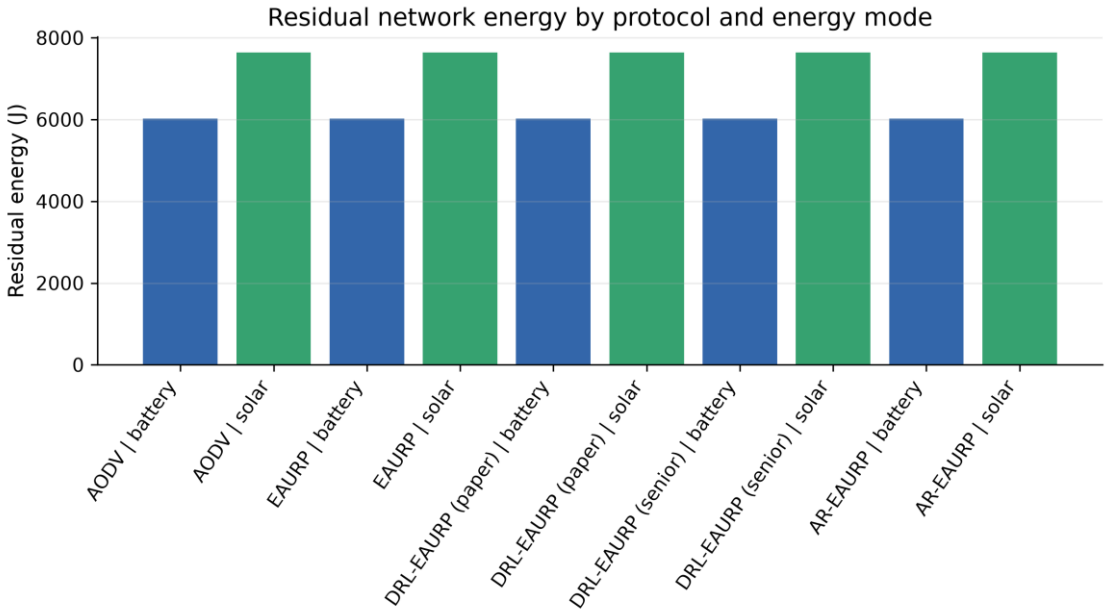


Figure 4. Total network residual energy under battery-only and solar/kinetic harvesting conditions.

In the 900 s lifetime experiment, all battery-only policies reached first death at about 423 s and 80% dead at about 425 s. AR first death changed from 423.60 ± 0.69 s to 432.79 ± 1.90 s with harvesting. The aggregate observation flag for the harvested 80%-dead endpoint is zero for every policy: none of those endpoints was observed in the five runs before the horizon. Those cases are right-censored at 900 s; they are not measured death times. Table 8 preserves the original censored marker. End-of-run residual energy and first-death times support an effect within the synthetic energy model, not a validated field-lifetime multiplier.

Table 8. Lifetime endpoints and total remaining energy. Harvested 80%-dead endpoints are censored in all five runs per policy.

Configuration	First death (s)	80% dead (s)	Residual energy (J)
AODV / battery	422.65 ± 0.56	424.55 ± 0.15	0.00 ± 0.00
AODV / solar	432.10 ± 2.28	censored / mixed	1395.71 ± 166.72
EAURP / battery	423.30 ± 0.58	424.90 ± 0.20	0.00 ± 0.00
EAURP / solar	432.84 ± 1.55	censored / mixed	1370.50 ± 190.10
DRL-EAURP (paper) / battery	423.50 ± 0.38	424.98 ± 0.33	0.00 ± 0.00
DRL-EAURP (paper) / solar	432.20 ± 2.51	censored / mixed	1402.79 ± 212.21
DRL-EAURP (senior) / battery	422.89 ± 0.58	424.76 ± 0.11	0.00 ± 0.00
DRL-EAURP (senior) / solar	432.12 ± 2.34	censored / mixed	1394.65 ± 167.13
AR-EAURP / battery	423.60 ± 0.69	425.12 ± 0.34	0.00 ± 0.00
AR-EAURP / solar	432.79 ± 1.90	censored / mixed	1360.29 ± 212.01

5.5 Ablations, forecast quality, and delivery-floor sensitivity

Table 9 shows the component-removal experiment at 30% gray holes. Full AR achieved 83.90 ± 8.08% PDR, versus 85.63 ± 8.04% without GAN and 84.50 ± 7.35% without CMDP. These point estimates do not establish that the complete stack outperforms its simpler variants. The energy-excuse removal matched the full configuration in the reported endpoints. This equality is a result for this workload, not evidence that the mechanism can never matter. Removing

components can change routes and hence the windows available to the detector, so differences are not independent causal contributions that can be added together.

Table 9. Component ablations. Disabled-detector zeros/default AUC are not classification results.

Configuration	PDR	Window TPR	Window FPR	Score AUC
C full	83.90 ± 8.08	7.82 ± 4.98	5.24 ± 5.14	0.55 ± 0.07
C without GAN	85.63 ± 8.04	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.00
C without LSTM	83.97 ± 7.25	8.89 ± 8.43	5.92 ± 3.67	0.56 ± 0.03
C without CMDP	84.50 ± 7.35	6.77 ± 8.57	5.55 ± 5.21	0.60 ± 0.09
C bare	82.67 ± 2.84	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.00
C without energy excuse	83.90 ± 8.08	7.82 ± 4.98	5.24 ± 5.14	0.55 ± 0.07

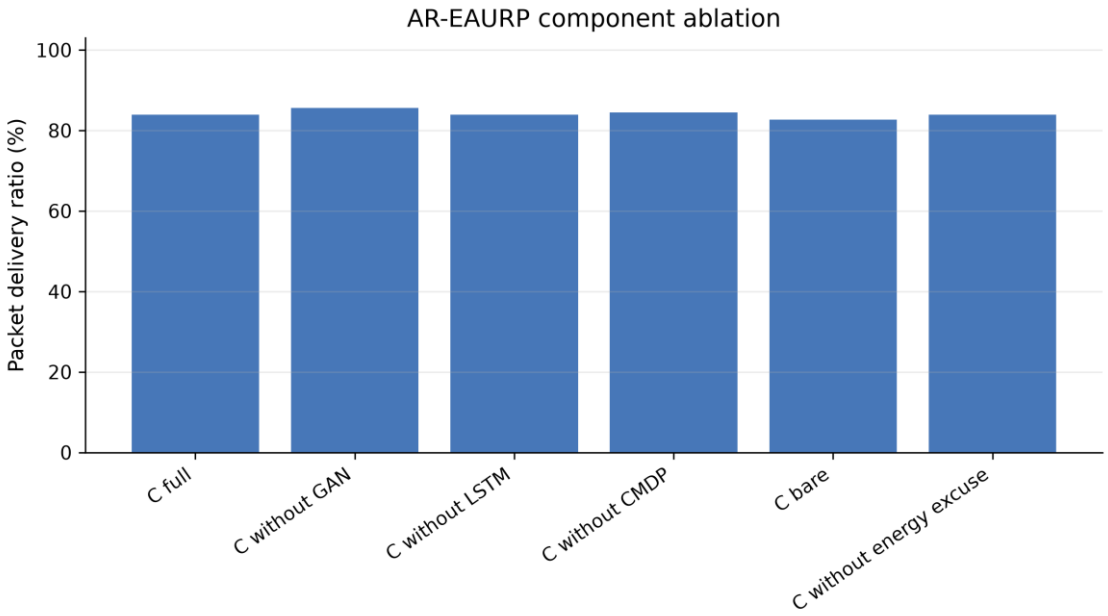


Figure 5. Component-ablation PDR. Point estimates and between-run spread do not show a uniform benefit from adding every module.

The forecaster provides a particularly important negative result. In the full 30% gray-hole configuration, LSTM holdout MAE was 0.495 ± 0.409 W, persistence MAE 0.057 ± 0.013 W, and window-mean MAE 0.095 ± 0.028 W. Persistence was selected in all five runs. Thus the experiment tests the fallback-equipped forecasting module, but it does not demonstrate useful LSTM prediction. The native system correctly avoids deploying the worse predictor under its selection rule. Improving model capacity without changing the short training horizon or validation design is not established as the solution.

Table 10 shows that delivery is not monotonic in the target floor. The 0.80 floor produced the highest descriptive PDR, 88.63 ± 4.07%; floors 0.90 and 0.95 produced 83.90 ± 8.08% and 78.20 ± 12.22%. Lambda increased with the stricter floors, showing that the penalty reacted, but a larger penalty did not ensure constraint satisfaction. These finite-run results identify a learning and feedback problem; they do not prove mathematical infeasibility of the underlying constrained routing problem. Choosing 0.80 after examining this sweep would require fresh confirmatory evaluation.

Table 10. Delivery-floor sensitivity and final/peak dual variable; n=5 per setting.

Configuration	PDR	Final CMDP λ	Peak CMDP λ
C floor=0.70	70.77 ± 20.20	0.43 ± 0.54	0.50 ± 0.50
C floor=0.80	88.63 ± 4.07	0.70 ± 0.65	0.86 ± 0.54
C floor=0.90	83.90 ± 8.08	1.69 ± 0.76	1.71 ± 0.72
C floor=0.95	78.20 ± 12.22	2.33 ± 0.99	2.33 ± 0.99

5.6 Payload protection and control overhead

At 30% gray holes, AR generated 218,395 ± 14,430 control bytes and 30,906 ± 1,356 cryptographic overhead bytes, with delay 0.28 ± 0.14 s. EAURP generated 224,339 ± 10,551 control bytes, about 30,654 ± 495 cryptographic bytes, and delay 0.27 ± 0.08 s. The no-protection AR setting recorded zero cryptographic bytes, PDR 84.03 ± 4.51%, and delay 0.24 ± 0.16 s. Table 11 reports all variants. Changing protection also changes packet sizes and subsequent network events; the observed delay difference is not an isolated measurement of cryptographic CPU latency. No claim is made about hardware execution time or formal security strength from these performance counters.

Table 11. Application PDR, aggregate protection/control bytes, and conditional delivered-packet delay.

Configuration	PDR	Crypto overhead (B)	Control bytes	Delay (s)
AODV	83.67 ± 9.62	30856.00 ± 1620.25	180906.20 ± 13723.64	0.21 ± 0.14
EAURP	82.67 ± 2.84	30654.40 ± 494.74	224338.60 ± 10550.78	0.27 ± 0.08
DRL-EAURP (paper)	82.60 ± 10.65	30699.20 ± 1761.51	223893.80 ± 31384.10	0.23 ± 0.10
DRL-EAURP (senior)	86.40 ± 6.85	31326.40 ± 1143.28	206686.40 ± 14286.77	0.21 ± 0.06
AR-EAURP	83.90 ± 8.08	30906.40 ± 1355.82	218395.00 ± 14430.38	0.28 ± 0.14
C without ECC	84.03 ± 4.51	0.00 ± 0.00	230747.60 ± 24773.70	0.24 ± 0.16

5.7 Separate thirty-pair commissioning follow-up

The commissioning-trained CMDP was additionally evaluated against EAURP with seed 1 and matched run numbers 21–50, at 24 nodes, six flows, and 25% gray-hole prevalence. This cohort is separate from E3. EAURP mean PDR was 79.81% and AR mean PDR was 80.97%. AR won 16 of 30 pairs. The paired estimate was +1.16 points with 95% CI −2.97 to +5.30. Mean delay changed from 0.256 s to 0.288 s; its paired difference was +32 ms with interval −21 to +84 ms. Neither endpoint establishes a reliable benefit.

Window TPR was 4.69%, FPR 4.94%, and AUC 0.586. Earlier exploratory candidates, including greedy and reduced-exploration routes, did not show a stable advantage. Those groups are retained in the companion report instead of selectively presenting the favorable small subset. The thirty-pair follow-up uses new run numbers relative to those earlier groups, but was selected during development in the same simulator. It is not preregistered independent external validation, and it cannot be pooled with the review cells as if all settings and implementation states were identical.

5.8 Earlier Python evidence and cross-model interpretation

The saved Python full-profile 30% gray-hole results gave AR PDR 50.25 ± 5.86% and EAURP PDR 47.92 ± 5.24%, a descriptive +2.32-point difference. The native E3 comparison instead gives +1.23 points with a broad paired interval. The Python context used 100 nodes, a larger area, discrete simulation rounds, and different communication and observation assumptions. Its results motivated native testing but cannot guarantee the same ordering under packet-level wireless behavior. A CSV establishes saved measurements, not that every artifact was executed in Colab; current runner notebooks are not proof of historical execution location.

Table 12. Historical Python 30% gray-hole slice, mean ± sample SD over five runs. Detector fields apply only where a comparable detector exists.

Native-label counterpart	Historical Python PDR (%)
AODV reference	46.47 ± 5.75
EAURP	47.92 ± 5.24
DRL-P	45.96 ± 6.01
DRL-S	46.41 ± 5.93
AR-EAURP	50.25 ± 5.86

AR's historical Python detector TPR was 29.33 ± 7.42% and FPR 1.43 ± 0.90%, also sample SD rather than confidence-interval half-widths. These differ substantially from native gray-hole detection. Absolute delivery values across

simulators should not be subtracted as a treatment effect. The strongest defensible statement is that the richer model exposes limitations that the earlier review did not resolve.

6. Discussion

6.1 Why detection and routing performance diverge

Successful detection is only one link in a longer chain. The relay must first obtain forwarding opportunities; a neighbor must observe enough evidence; a detector must score it correctly; evidence must meet consensus rules; the route strategy must then find a viable alternative before excessive loss occurs. A black hole may be easy to identify after dropping many packets while still leaving the network with no good path. A gray hole can produce too few selective drops in an eligible window to separate clearly from benign wireless loss. The data support these limitations without identifying one mechanism as the sole cause.

This distinction also explains why accuracy alone is an inadequate headline result. A strong TPR with a large FPR can exclude useful relays. A high PDR can coexist with low attack detection when the attack rarely lies on active routes. A low conditional delay can reflect severe loss. Joint reporting of PDR, latency, detector behavior, consensus state, energy, and control overhead is therefore more informative than selecting a single favorable metric.

6.2 Implications for model complexity

The native results do not justify the assumption that adding GAN, LSTM, and CMDP components must outperform the substrate. Persistence outperformed the LSTM, full AR did not dominate its ablations, and increasing the delivery floor did not monotonically improve PDR. The implementation's fallback rule is useful engineering, but its use is a reason to qualify the role of the neural forecaster. Simpler anomaly baselines and nonlearning route policies should receive the same held-out evaluation before complexity is accepted as beneficial.

The shared CMDP also receives a nonstationary stream of network outcomes while actions affect future observations. Delayed timeout feedback, exploration, and finite training can all alter the state and penalty trajectory. Commissioning gives the agent prior clean experience, yet the thirty-pair interval remains inconclusive. This suggests that additional replications alone are not a guaranteed route to favorable results; improved measurement, baseline selection, and policy design may be necessary.

6.3 Threats to validity

Internal validity is limited by shared model parameters, reused clean threshold-calibration data, continuing evaluation-time learning, and protocol-dependent random-stream consumption. Comparator variants share a custom harness, which improves implementation consistency but does not establish equivalence to independently released protocol implementations. Cryptographic CPU time and model distribution cost are not isolated. The source/build identity records the reviewed checkpoint lineage rather than proving correctness by itself.

Statistical validity is limited by five runs per complete review condition, multiple unadjusted contrasts, dependent windows within a run, and incomplete high-density replication. Wide intervals permit both practically useful gains and meaningful harm. Absence of a significant difference is not evidence of equivalence. The isolated trust-poisoning interval is exploratory. No claim of exhaustive parameter coverage, preregistration, or a powered confirmatory superiority test is made.

External validity is limited by one radio configuration, one synthetic harvesting process, random waypoint mobility, a modest base network, selected packet sizes, and specific attacker rules. No real hardware multi-hop benchmark or calibrated outdoor deployment is included. The separate RelayMesh direct-message application is a project artifact, not experimental proof of AR-EAURP routing. These boundaries are necessary when translating a final-review demonstration into a research claim.

7. Reproducibility and artifact availability

The native implementation is under ns3/contrib/ar-eaurp. The review evidence is under results/ns3-native-paper-review, including study_settings.json, configuration_settings.csv, run_settings.csv, per-experiment aggregate CSVs, paired_e3_summary.json, and pairwise_e3_comparisons.csv. The per-run ledger identifies 863 included checkpoints and their hashes. The source/build SHA-256 is 45da34216e04a79d26b8e9db9e02270a2bcdb5c3edf8e63fea4d61eda8d9ed02. The original root manifest selection predates the composite E1–E6 report; the run ledger defines its actual scope.

The experiment runner validates checkpoint contents, settings, and source identity before resuming. The review command is provided in Appendix A. Running it to completion would fill missing work and create a later evidence state; it does not retroactively change the counts reported here. The separate commissioning follow-up is stored under results/optimization_study, with settings and adjacent logs for every JSON result. Historical Python aggregates are under output/full_results/csv. The companion Test Results and Findings report retains all 180 E3 pairwise contrasts, exploratory groups, engineering checks, and detailed artifact references.

NetAnim XML is generated by ns-3's animation instrumentation during a simulation run. Replaying that XML visualizes recorded events; it does not rerun routing or change the parameters that produced the trace. Live runs and trace replay are separate operations. A browser animation is an explanatory aid, while the JSON/CSV checkpoints and settings ledgers provide the quantitative evidence used here. A packet-event animation alone cannot establish a protocol's comparative performance.

8. Conclusions and future work

AR-EAURP has been implemented as a native ns-3 routing system with explicit forwarding attacks, observable watchdog evidence, clean commissioning, forecast validation, constrained route learning, and protected application traffic. The work advances the earlier Python review by exposing radio, routing, observation, and energy interactions and by preserving transparent settings, checkpoints, variability, and paired comparisons.

The available evidence is mixed. Native gray-hole detection is weak, the learned forecaster loses to persistence in the selected workload, and the complete stack does not consistently outperform simpler policies or ablations. One unadjusted attack contrast favors AR, but most intervals and the separate thirty-pair follow-up do not establish a reliable PDR gain. Harvesting improves modeled energy availability, while lifetime claims remain bounded by the synthetic process and censoring. These are findings to report, rather than outcomes to tune until a preferred ranking appears.

Future work should complete the deferred density replications, introduce independent clean calibration and deployment-relevant attack validation, compare simple detector alternatives, assess a frozen decentralized policy, and calibrate energy and channel models. A subsequent confirmatory study should define a meaningful effect size, plan replication and multiplicity control, and reserve independent evaluation conditions before tuning. The present contribution is a reproducible implementation and a detailed empirical account of both its capabilities and its unresolved limitations.

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Appendix A. Reproduction procedure and reporting conventions

From the project root in WSL, the existing native study entry point is:

```
bash ns3/scripts/run.sh --profile=review --only=e1,e2,e3,e4,e5,e6 --out=results/ns3-native-paper-review --render=0
```

The separate commissioning comparison can be reproduced with:

```
export NS3_HOME="$HOME/.cache/ar-eaurp/ns-3.45"
bash ns3/scripts/run_cmdp_commissioning_study.sh results/optimization_study/reproduction 21 50
```

These are reproduction instructions, not a statement that new simulations were executed during manuscript preparation. The installed ns-3 checkout and supporting setup must match the native module. Mean ± SD is used throughout unless a CI is explicitly labeled. PDR differences are percentage points, not relative percent changes. Missing observations remain absent, censored endpoints remain censored, and disabled detectors remain not applicable.

AR-EAURP: Experimental Methods, Complete Test Results, and Findings

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Research objectives, experimental logic, and decision rules

This report answers four practical research questions: how each native policy behaves across mobility, density, attacks, and energy modes; whether anomaly evidence becomes successful routing protection; whether individual components justify their complexity; and whether matched AR–EAURP differences are stable enough to support a performance claim. It is an experimental record, not an assertion that all results favor the proposed method. The appropriate unit of replication is a simulator run, not an individual packet or repeated detector window.

The native C++ study is the principal current implementation evidence. Historical Python results provide the earlier-review context. The earlier native five-run snapshot, exploratory optimization groups, and later thirty-pair commissioning study are distinct cohorts. They use different run ranges and, in some cases, implementation states. Their sample sizes are not pooled, and a change between cohorts is not interpreted as a controlled estimate of a code change's effect.

A result is described as a point advantage when the mean difference is positive. An interval spanning zero is inconclusive for direction. A nonzero exploratory interval is not automatically a confirmatory finding, particularly across eighteen selected contrasts and 180 all-pair comparisons. Missing runs are not filled with averages or zeros. A single run has no empirical SD. A disabled detector's zero/default outputs are not evidence of perfect classification. A censored lifetime is not an observed death time. These conventions apply to every table below.

Native implementation and complete measurement definitions

Network and routing substrate

The system consists of mobile wireless nodes that originate application traffic, discover routes, and relay packets over an IEEE 802.11 ad hoc network. Each node maintains neighbor information, residual energy, routing state, and locally available forwarding observations. Packets traverse actual ns-3 IP, MAC, and PHY events. Delivery therefore depends on route discovery, queueing, retransmission, interference, mobility, relay behavior, and battery state. A packet is not declared delivered by a single network-wide probability. This distinction is central to the transition from the earlier Python study to the native experiment.

The implementation is a native C++ ns-3.45 module. The detector, forecaster, neural-network operations, tabular agents, and constrained route selector execute within that module. Python scripts aggregate saved outcomes and prepare reports; they do not replace the packet path with an external predictor. The protocol uses route requests, route replies, route errors, neighbor advertisements, trust reports, and revocation-related control messages. A route can fail before an attacker is reached, so malicious-drop counters and total application loss measure different processes.

For a neighbor j, forwarding trust is smoothed as $T_j^{new} = 0.7T_j^{old} + 0.3PFR_j$, where PFR is the observed forwarding ratio. The trust and energy route score has the form $R_j = \mathrm{clip}(a_i T_j + b_i E_j^{eff}, 0,1)$. The adaptive energy weight uses the mean normalized energy of the node and its nonexpired neighbor advertisements; the trust weight is its complement. The link cost includes $(1 - R_j)$ and a 0.35 hop penalty, represented internally in integer milli-cost units. This is a routing heuristic rather than a proof of an optimal path under the radio model.

Energy-aware variants exclude relays below 20% of their initial charge. The AODV-style reference uses equal hop costs and does not apply that energy gate. A contested AR neighbor receives an additional 0.5 trust multiplier for a 60 s contested interval. These choices can improve avoidance but also remove useful relays. Their possible cost is measured through PDR, delay, route failures, and component ablations.

Threat model

Attacks are insider forwarding and reputation attacks. Malicious nodes participate in the network and can forward control traffic while withholding relayed application packets. A black-hole relay drops relayed data. A gray-hole relay forwards smaller packets and independently drops eligible larger packets with probability 0.25, using a 700-byte size threshold. The trust-poisoning configuration combines selective forwarding with dishonest reports: a colluder advertises favorable trust for fellow malicious nodes and low trust for honest nodes. The explicit collusion assumption allows an attacker to recognize its allies. Honest routing does not receive that ground-truth attacker list as an avoidance policy.

The experiment varies the configured hostile fraction from 0% to 50% in ten-percentage-point steps. With a finite number of nodes, configured prevalence is not a guarantee that every route encounters an attacker or that every adversarial node obtains a forwarding opportunity. Selective packet eligibility, sparse observations, and route changes further separate attack prevalence from the number of positive detector windows. Consequently, a 30% gray-hole condition is not equivalent to dropping 30% of all generated packets.

The evaluated scope excludes physical jamming, malicious radio hardware, cryptographic key theft, wormholes, and arbitrary compromise of the simulator itself. A low delivery ratio can still arise from benign link failure or a depleted

relay. These competing explanations motivate careful watchdog accounting and an energy excuse before trust condemnation. No formal anonymity or traffic-unobservability property is established by the experiments.

Watchdog observations and detection layers

The watchdog associates a forwarded packet digest with the intended next hop and confirms an overheard onward transmission using the neighbor MAC binding. A failed link-layer handoff cancels its pending watchdog entry, avoiding the automatic attribution of that failed handoff to malicious forwarding. A missing observation is therefore handled differently from a confirmed successful forwarding event. The watchdog timeout is 1 s. Published opinions require sufficient accumulated evidence, and an anomaly window requires at least six new observations.

The native detector consumes nine features: short-window PFR, long-window PFR, their difference, large-packet drop ratio, small-packet drop ratio, scaled mean forwarding latency, scaled latency standard deviation, report disagreement, and the difference between large- and small-packet drop ratios. Latencies are scaled by 0.1 s and capped at two; features are sanitized and clipped to the implemented range. Size-conditioned observations target selective forwarding that can remain hidden in a single average PFR.

Three measurement layers are retained. First, window-level TPR, FPR, and AUC describe thresholded anomaly scores and score ranking on eligible windows. Second, contested-node measurements describe corroborated suspicion with honest-witness requirements. Third, revocation measurements describe the protocol's final exclusion state. A correct anomaly score is not automatically a corroborated accusation, and an accusation is not automatically successful rerouting. The consensus logic requires at least two accusers and at least half of the relevant observers, subject to local evidence and energy checks. Ground-truth labels support evaluation; they are not a substitute for this inference path.

Clean-trained generative anomaly model

The detector is trained during a separate 240 s attack-free commissioning simulation. For feature k, clean training statistics define $z_k = (x_k - \mu_k)/\max(\sigma_k, 10^{-3})$. At least 32 clean windows are required. The generator maps an eight-dimensional latent vector through two 32-unit ReLU layers to nine outputs. The discriminator uses layers of 32 and 16 units with leaky ReLU activations and a sigmoid output. Adversarial training uses 120 epochs, batches of 16, a learning rate of 2×10^{-4}, and one-sided real-label smoothing to 0.9. These are native implementation settings, not parameters copied from the older Python detector.

Following training, the generator produces a bank B of 2,048 standardized normal behavior vectors. The inference score is the nearest generated-vector root-mean-square distance:

$$s(x) = \min_{b \in B} \sqrt{\frac{1}{9}\sum_{k=1}^{9} (z_k - b_k)^2}.$$

The threshold is the 95th percentile of scores on the collected clean training windows. The score is not a maliciousness probability and is not the discriminator's real-versus-generated output. The model is related to generative anomaly modeling, but the generated-bank inference procedure is specific to this project. In particular, it does not optimize a latent vector separately for every evaluation sample.

Detector parameters are frozen for attack evaluation. This separates attack observations from model fitting, but the same clean collection is used for fitting and threshold calibration. The threshold therefore lacks an independent clean calibration set, and nominal 5% calibration does not guarantee a 5% FPR on new network conditions. Mobility, neighborhood changes, sparse evidence, and channel-induced losses can alter the score distribution. The measured evaluation FPR is reported instead of assuming the calibration target was achieved.

Harvest forecasting and fallback selection

The forecaster predicts mean harvested power over the next ten samples from a history of twenty samples. At each 1 s sample, the inputs are normalized harvested power and the sine and cosine of the modeled day phase. A 24-unit LSTM is trained for 30 epochs with learning rate 0.005. Its target is power in watts, not cumulative battery energy. Routing blends current normalized energy and normalized predicted harvested power for its energy-oriented action.

For each node's commissioning trace, the earliest 70% supplies training data and the last 30% supplies validation. Sequences crossing the split are purged, and normalization uses training power only. Persistence and a recent-window mean are evaluated on the same held-out targets. The LSTM is used for routing only when its validation MAE is strictly below persistence; otherwise prediction falls back to persistence. The field historically named seasonal MAE corresponds to a window-mean predictor, so this paper does not call it a seasonal forecasting baseline.

All traces contribute to shared model parameters. There is one fleet-trained detector and one fleet-trained forecaster, rather than a separately trained network at each node. Provisioning such models is a plausible deployment step, but training communication, model distribution, and hardware inference cost are not evaluated as a distributed field system. Reporting the fallback outcome is necessary: an experiment in which persistence is selected cannot be cited as evidence that the LSTM improved delivery.

Constrained learning for route strategy selection

The CMDP-inspired agent chooses among trust-oriented, energy-surplus, balanced, and diversified routing strategies. Its six state variables are mean local trust, own residual-energy fraction, normalized speed, forecast surplus, contested-neighbor fraction, and rolling application feedback PDR. The feedback window contains the latest 100 delivery outcomes. Before feedback exists, its initial PDR value is one. This online state differs from the final measured application PDR and can be affected by timeout feedback.

The route selector uses a six-input, two-hidden-layer network with 64 ReLU units per layer and four action values. Its Double-DQN target is

$$y = r + 0.9Q_{target}(s', \operatorname{argmax}_a Q_{online}(s', a)).$$

The shaped reward is

$$r = (2I_{delivered} - 1) - 0.05c_E - 5\lambda \max(0, \tau - \widehat{PDR}_{100}).$$

Here c_E is a hop-based energy proxy, and τ is the target delivery floor. The proxy derives from a 2 mJ-per-hop assumption and internal scaling; it is not the radio energy consumed by the packet. Actual battery behavior is measured separately through the ns-3 energy model. Every ten source attempts, the dual variable is updated as $\lambda \leftarrow \text{clip}(\lambda + 0.05(\tau - \widehat{PDR}_{100}),0,20)$.

The replay buffer holds 5,000 transitions; batch size is 32, learning rate is 0.001, and training is attempted every four observations. Target parameters use a hard copy every 200 training calls and a 0.01 Polyak update otherwise. Epsilon decays from 0.30 toward 0.05 with an 800-decision time constant. Exploration and shared policy updates continue during evaluation. The current implementation also learns from clean commissioning outcomes, rather than first activating the route learner under attack.

One CMDP agent is shared through the native experiment coordinator. This parameter sharing is a substantive simulation assumption: the study evaluates a coordinated fleet policy, not independently learning decentralized agents. The Lagrangian construction is motivated by constrained decision processes, but no convergence, constraint-feasibility, or safety guarantee follows from the implemented finite-run learner. The floor sweep explicitly measures whether higher targets translate into higher observed delivery.

Baseline definitions and protected payloads

All five comparison policies share the native packet harness. AODV denotes the custom equal-hop AODV-style reference; it is not ns-3's stock AODV implementation. A separate ns3-aodv integration option exists for clean checks but is not included in the attack matrix. EAURP is the project's trust/energy substrate adaptation, inspired by the base protocol. DRL-P and DRL-S denote native adaptations of the source-equation and delivered-implementation variants. Their tabular agent discretizes state, uses two actions, learning rate 0.1, discount 0.9, epsilon 0.1, and actual delivery feedback. Their behavior must not be described as verbatim reproduction of an external publication's complete original implementation.

Protected application traffic uses X25519 key agreement, HKDF-SHA256 key derivation, and AES-256-GCM authenticated encryption. A 12-byte nonce and 16-byte tag add 28 bytes per protected payload. The measured aggregate overhead includes the packets counted by the native instrumentation; it is not simply the number of measured source packets multiplied by 28. Key provisioning is trusted in the simulation. Payload protection does not prove authentication of every routing advertisement or traffic-analysis resistance. The no-ECC label in saved outputs disables the implemented payload protection path and should be interpreted in that operational sense.

Metric dictionary and denominator audit

Endpoint	Definition, unit, and interpretation
PDR	Unique received application packets / measured generated packets; fraction in raw data, percentage in tables
Mean delay	Delivered-packet arrival minus creation time, seconds; conditional on delivery

Endpoint	Definition, unit, and interpretation
Throughput	Received application bytes × 8 / measurement_seconds, bit/s
Source feedback PDR	Rolling latest 100 success/timeout outcomes; learner state, not final application PDR
Detector TPR/FPR	Thresholded score outcomes over eligible positive/negative windows; not independent packet trials
Detector AUC	Ranking of window scores when both classes exist; default values without a class are not validation
Contested TPR/FPR	Native corroboration-state measures, separate from anomaly windows
Revocation TPR/FPR	Final exclusion-state measures, including ordinary trust logic
Residual energy	Sum of remaining network battery energy, J; not per-node charge
First death	Observed time to first dead node, s; use corresponding observation flag
80% dead	Time to at least 80% dead; unobserved endpoints are right-censored
Control bytes	Aggregate native routing-control accounting, B; not application goodput
Crypto overhead	Aggregate protection bytes counted by instrumentation, B; no-ECC disables payload protection
Forecast MAE	Mean absolute power-prediction error on purged chronological holdout, W
Lambda	Nonnegative bounded learner penalty; increasing lambda is not proof that a constraint is satisfied

Statistical procedure and repeated clean conditions

For n completed run values, the sample SD uses n−1 in the denominator. The 95% t interval is mean ± t(0.975,n−1) × SD / sqrt(n). For five runs the critical value is approximately 2.776. Rate intervals in the saved report are bounded to [0,1]; this bounding does not make them exact binomial intervals. A paired PDR difference is 100 × (PDR_A − PDR_B) within a matched run number and is summarized across the five differences. The interval is calculated from their SD, not from the SD of either protocol alone.

The review master seed is 20260909 and run numbers are 1–5. Thus the study uses distinct run streams under one seed, not five distinct master seeds. Matching run numbers does not force identical random draws after protocols take different event paths. Zero-hostile rows under three attack labels describe the same clean setting and must not be counted as three independent confirmations. The 180-contrast appendix is exhaustive for the recorded E3 pairwise family but not for every possible network parameter combination.

Detailed completion and missing-data explanation

E2 contains 100 completed runs at 60, 80, 100, and 120 nodes (five policies × five runs × four densities), five single runs at 150 nodes, and three single runs at 200 nodes. EAURP and AR have no 200-node checkpoint in the included evidence. The missing 42 runs therefore consist of the remaining replications at 150/200 nodes plus the two absent 200-node policy conditions. All other review experiments meet their planned five-run budget. No new runs were started during this document revision.

The aggregate settings key named measurement_duration_s=120 is a historical label. Raw measurement_seconds=100 is the actual base measurement denominator after the 20 s warm-up. The 900 s lifetime horizon has 880 s of measured generation. These distinctions should be preserved when quoting throughput, packet counts, commissioning cost, or experiment duration.

Scope and evidence inventory

This document records the saved test results and the findings available on 22 October 2026. It accompanies the research manuscript and contains the detailed native ns-3 experiment tables, simulation settings, uncertainty estimates, all 180 E3 pairwise PDR contrasts, the separate CMDP commissioning study, exploratory tuning groups, an earlier native validation snapshot, and historical Python results.

The main native review matrix contains **179/181 conditions and 863/905 planned runs**. The remaining 42 density replications were deferred. Missing results are not imputed. The 30 matched CMDP commissioning pairs are a separate study with a different seed/run range, not an extension of the matrix's sample size. All numbers below come from saved artifacts; no additional simulation was run while preparing this document.

The evidence does not establish general AR-EAURP superiority. A positive PDR point estimate and a confidence interval spanning zero are reported as inconclusive. The single exploratory E3 interval excluding zero is unadjusted for multiple comparisons.

Author and provenance note: The five project authors are listed as provided. Institutional affiliation was not supplied and has not been invented. This record covers saved evidence as of 22 October 2026.

Timing clarification: the base native parameter `time=120` is the scenario duration, including the 20 s warm-up. The saved simulator metric `measurement_seconds` is 100 s; six 1-packet/s flows therefore send 600 measured application packets. Cooldown adds a further 10 s to allow final arrivals. Some earlier prose calls this a "120 s evaluation"; the exact metric window is 100 s, as recorded in the raw checkpoints. Clean commissioning is separate.

Executive summary

This report summarizes the native ns-3.45 review matrix within the available time. The planned budget is five run numbers per condition with a fixed ns-3 seed of 20260909; the completion table records actual coverage, and incomplete replications are not imputed. The E3 attack sweep includes a paired AR-EAURP versus EAURP PDR analysis from the checksummed per-run checkpoints.

Methods and reproducibility

Reproduction command from the project root: `bash ns3/scripts/run.sh --profile=review --only=e1,e2,e3,e4,e5,e6 --out=results/ns3-native-paper-review --render=0`. The experiment runner resumes only checksummed runs whose full parameter map and source/build identity match. Run `python generate_paper_report.py` from the result directory to rebuild the expanded settings ledgers and charts.

Execution provenance: this final E1–E6 report is assembled from verified native ns-3 run checkpoints. The root `manifest.json` selection field records the original C++ batch selection (`e3,e4,e5,e6`); E1 and the time-boxed E2 results were included from their verified checkpoint sets. The per-run ledger is the authoritative record of the 863 included runs.

Simulation settings

Each row in `configuration_settings.csv` names the canonical run-1 CLI arguments and aggregate metrics for one condition, including the actual replicate count and run numbers. Each row in `run_settings.csv` records the exact CLI argument map for that completed run, simulator-reported cfg_* values, selected per-run metrics, and the SHA-256 of the checksummed raw checkpoint. Confidence intervals are omitted where fewer than two replications exist.

Experiment	Factors varied	Conditions	Native runs
E1 mobility	5 protocols × max speed 1, 5, 10, 15, 20 m/s	25	125
E2 density	5 protocols × 60, 80, 100, 120, 150, 200 nodes	30	150
E3 attacks	5 protocols × 3 attacks × 0–50% hostile nodes (10-point steps)	90	450
E4 energy	5 protocols × battery/solar-kinetic harvesting	10	50
E4b lifetime	5 protocols × battery/solar-kinetic harvesting, 900 s horizon	10	50
E5 ablation	6 AR-EAURP component configurations	6	30
E5b CMDP floor	0.70, 0.80, 0.90, 0.95 PDR constraints	4	20
E6 security overhead	5 protocols plus AR-EAURP without ECC	6	30

- Simulator: ns-3.45, native C++ source/build identity SHA-256 `45da34216e04a79d26b8e9db9e02270a2bcdb5c3edf8e63fea4d61eda8d9ed02`.
- Protocols: AODV, EAURP, DRL-EAURP (paper), DRL-EAURP (senior), AR-EAURP.
- Base review scenario: 24 nodes, six CBR flows, 420 m square area, 120 s scenario including 20 s warm-up (100 s measured), 10 s cooldown.

- AR-EAURP clean commissioning: 240 s; GAN 120 epochs; LSTM 30 epochs. The CMDP route selector also learns during clean commissioning before attack evaluation.
- RNG: fixed seed 20260909 and run numbers 1–5 planned per condition. Run numbers are matched between protocol comparisons; results are stochastic replications, not repeated packets from one run.
- Metrics are reported as mean ± sample SD when at least two replications are present. Per-condition 95% confidence intervals use a two-sided Student t interval with degrees of freedom based on the observed sample size; bounded rate intervals (PDR, TPR, FPR, and AUC) are constrained to [0, 1]. E3 primary paired differences compare AR-EAURP and EAURP using corresponding run numbers.
- All ten pairwise protocol contrasts are retained for each E3 attack/prevalence condition in `pairwise_e3_comparisons.csv`; their intervals are unadjusted and should be interpreted as exploratory across this multiple-comparison family.
- `configuration_settings.csv` records each condition's canonical run-1 CLI parameter map, observed run numbers, aggregate metrics, and simulator-reported settings. `run_settings.csv` records the exact CLI map for every completed run, measured cfg_* values, seed/run number, and the checksum of its raw checkpoint. Aggregate tables in `csv/` retain cfg_* means and sample SDs.

The review profile retains the Colab notebook's E1–E6 factor levels but uses a 120 s scenario (100 s measured) and the 24-node/six-flow review workload rather than the notebook full profile's 400 s and larger default workload. The planned budget is five runs. Any time-boxed shortfall appears in the completion table; this is not a byte-for-byte rerun of the Colab full-profile configuration.

Completion status

Completed 179 of 181 planned conditions and 863 of 905 planned native runs. The time-boxed E2 density sweep has missing replications; they are absent from the ledgers and are not imputed.

Experiment	Completed conditions	Completed runs
e1_speed	25/25	125/125
e2_density	28/30	108/150
e3_attack	90/90	450/450
e4_energy	10/10	50/50
e4b_lifetime	10/10	50/50
e5_ablation	6/6	30/30
e5b_cmdp_floor	4/4	20/20
e6_overhead	6/6	30/30

E1 — Mobility

Configuration	PDR	Delay (s)	Throughput (bit/s)
AODV @ 1 m/s	94.53 ± 4.92	0.12 ± 0.10	34932.50 ± 1773.64
AODV @ 5 m/s	92.57 ± 3.41	0.12 ± 0.08	34212.93 ± 1181.96
AODV @ 10 m/s	90.13 ± 5.59	0.21 ± 0.10	33258.11 ± 2130.37
AODV @ 15 m/s	91.47 ± 2.22	0.21 ± 0.05	33777.76 ± 747.07
AODV @ 20 m/s	89.17 ± 5.34	0.24 ± 0.14	32930.64 ± 1999.54
EAURP @ 1 m/s	86.80 ± 9.64	0.10 ± 0.10	32071.52 ± 3523.84
EAURP @ 5 m/s	95.00 ± 4.99	0.18 ± 0.15	35081.84 ± 1894.53
EAURP @ 10 m/s	88.10 ± 7.01	0.23 ± 0.12	32486.45 ± 2707.45
EAURP @ 15 m/s	88.90 ± 6.38	0.30 ± 0.10	32870.02 ± 2304.32
EAURP @ 20 m/s	87.07 ± 6.41	0.29 ± 0.10	32138.88 ± 2366.01
DRL-EAURP (paper) @ 1 m/s	93.87 ± 5.78	0.09 ± 0.02	34676.83 ± 2041.94
DRL-EAURP (paper) @ 5 m/s	90.70 ± 8.69	0.19 ± 0.19	33476.00 ± 3202.31
DRL-EAURP (paper) @ 10 m/s	89.97 ± 5.21	0.24 ± 0.13	33172.64 ± 1953.22

Configuration	PDR	Delay (s)	Throughput (bit/s)
DRL-EAURP (paper) @ 15 m/s	90.97 ± 4.89	0.30 ± 0.08	33570.30 ± 1859.21
DRL-EAURP (paper) @ 20 m/s	85.67 ± 8.71	0.28 ± 0.13	31650.02 ± 3159.78
DRL-EAURP (senior) @ 1 m/s	96.30 ± 3.57	0.12 ± 0.08	35571.57 ± 1307.86
DRL-EAURP (senior) @ 5 m/s	95.10 ± 4.18	0.14 ± 0.09	35086.50 ± 1611.90
DRL-EAURP (senior) @ 10 m/s	91.63 ± 3.48	0.19 ± 0.07	33805.54 ± 1327.14
DRL-EAURP (senior) @ 15 m/s	91.23 ± 3.54	0.22 ± 0.11	33687.44 ± 1365.07
DRL-EAURP (senior) @ 20 m/s	92.20 ± 2.78	0.28 ± 0.08	34059.62 ± 1042.92
AR-EAURP @ 1 m/s	86.60 ± 14.32	0.12 ± 0.14	31991.66 ± 5181.71
AR-EAURP @ 5 m/s	94.10 ± 5.56	0.19 ± 0.12	34783.52 ± 2094.29
AR-EAURP @ 10 m/s	83.93 ± 11.10	0.22 ± 0.09	30954.80 ± 4053.12
AR-EAURP @ 15 m/s	86.17 ± 6.90	0.31 ± 0.13	31847.71 ± 2513.57
AR-EAURP @ 20 m/s	89.23 ± 4.73	0.27 ± 0.06	32886.26 ± 1742.36

E2 — Network density

Run count is shown for each condition. A single run is descriptive only and is not a replicated estimate; the settings ledgers show exact coverage.

Configuration	Runs	PDR	Delay (s)	Throughput (bit/s)
AODV (reference baseline) @ N=60	5	81.93 ± 5.71	0.25 ± 0.13	30248.45 ± 2179.59
EAURP (base substrate) @ N=60	5	89.47 ± 4.02	0.26 ± 0.06	32970.30 ± 1558.44
A: DRL-EAURP (paper) @ N=60	5	92.00 ± 5.39	0.28 ± 0.04	33989.07 ± 1885.60
B: DRL-EAURP (senior) @ N=60	5	91.40 ± 4.55	0.26 ± 0.05	33710.98 ± 1673.22
C: AR-EAURP (advancement) @ N=60	5	92.40 ± 4.79	0.23 ± 0.07	34144.48 ± 1668.36
AODV (reference baseline) @ N=80	5	77.43 ± 3.85	0.22 ± 0.07	28616.26 ± 1519.62
EAURP (base substrate) @ N=80	5	87.03 ± 2.72	0.23 ± 0.03	32123.42 ± 1079.28
A: DRL-EAURP (paper) @ N=80	5	83.60 ± 4.86	0.29 ± 0.06	30875.17 ± 1856.13
B: DRL-EAURP (senior) @ N=80	5	89.53 ± 3.11	0.28 ± 0.06	33080.03 ± 1123.74
C: AR-EAURP (advancement) @ N=80	5	88.00 ± 5.39	0.25 ± 0.05	32492.42 ± 1952.84
AODV (reference baseline) @ N=100	5	64.60 ± 11.08	0.25 ± 0.01	23852.34 ± 4045.73
EAURP (base substrate) @ N=100	5	70.00 ± 11.94	0.31 ± 0.07	25908.19 ± 4365.27
A: DRL-EAURP (paper) @ N=100	5	75.40 ± 6.80	0.40 ± 0.11	27885.86 ± 2431.58
B: DRL-EAURP (senior) @ N=100	5	74.23 ± 8.51	0.29 ± 0.04	27448.06 ± 3083.48
C: AR-EAURP (advancement) @ N=100	5	67.87 ± 11.27	0.26 ± 0.07	25089.18 ± 4112.41
AODV (reference baseline) @ N=120	5	56.90 ± 18.69	0.36 ± 0.16	21015.57 ± 6915.42
EAURP (base substrate) @ N=120	5	65.50 ± 19.00	0.29 ± 0.10	24207.65 ± 6917.94
A: DRL-EAURP (paper) @ N=120	5	59.20 ± 19.86	0.28 ± 0.14	21868.96 ± 7269.89
B: DRL-EAURP (senior) @ N=120	5	59.83 ± 19.34	0.33 ± 0.17	22168.42 ± 7095.11
C: AR-EAURP (advancement) @ N=120	5	60.70 ± 19.30	0.39 ± 0.20	22446.27 ± 7069.64
AODV (reference baseline) @ N=150	1	38.67 (n=1)	0.32 (n=1)	14312.80 (n=1)
EAURP (base substrate) @ N=150	1	39.83 (n=1)	0.35 (n=1)	14856.24 (n=1)
A: DRL-EAURP (paper) @ N=150	1	41.17 (n=1)	0.32 (n=1)	15351.44 (n=1)
B: DRL-EAURP (senior) @ N=150	1	41.67 (n=1)	0.38 (n=1)	15483.04 (n=1)
C: AR-EAURP (advancement) @ N=150	1	34.17 (n=1)	0.22 (n=1)	12717.20 (n=1)
AODV (reference baseline) @ N=200	1	28.50 (n=1)	0.12 (n=1)	10568.80 (n=1)
A: DRL-EAURP (paper) @ N=200	1	34.33 (n=1)	0.16 (n=1)	12732.72 (n=1)
B: DRL-EAURP (senior) @ N=200	1	32.50 (n=1)	0.21 (n=1)	12058.00 (n=1)

E3 — Attack prevalence by protocol

PDR cells show mean ± sample SD across five run numbers; the table keeps attack types and hostile-node percentages separate.

Blackhole

Hostile nodes	AODV	EAURP	DRL-EAURP (paper)	DRL-EAURP (senior)	AR-EAURP
0%	89.17 ± 5.34	87.07 ± 6.41	85.67 ± 8.71	92.20 ± 2.78	89.23 ± 4.73
10%	44.80 ± 3.62	71.33 ± 15.58	71.67 ± 13.16	41.27 ± 10.03	63.67 ± 16.29
20%	28.57 ± 9.14	45.80 ± 23.59	50.17 ± 14.74	34.93 ± 9.25	45.47 ± 18.08
30%	30.37 ± 6.55	40.80 ± 15.61	38.40 ± 14.38	25.97 ± 9.38	38.47 ± 15.13
40%	23.13 ± 8.54	30.33 ± 15.43	29.30 ± 15.53	24.20 ± 12.06	33.77 ± 10.16
50%	24.80 ± 11.27	20.53 ± 9.06	25.07 ± 9.73	21.77 ± 11.73	21.83 ± 7.74

Grayhole

Hostile nodes	AODV	EAURP	DRL-EAURP (paper)	DRL-EAURP (senior)	AR-EAURP
0%	89.17 ± 5.34	87.07 ± 6.41	85.67 ± 8.71	92.20 ± 2.78	89.23 ± 4.73
10%	87.10 ± 7.19	90.53 ± 5.05	89.50 ± 4.62	90.83 ± 4.44	87.43 ± 7.86
20%	84.27 ± 5.43	80.70 ± 8.88	84.63 ± 5.76	89.30 ± 5.79	87.37 ± 6.72
30%	83.67 ± 9.62	82.67 ± 2.84	82.60 ± 10.65	86.40 ± 6.85	83.90 ± 8.08
40%	82.50 ± 7.31	83.33 ± 8.92	83.03 ± 5.88	85.23 ± 7.72	81.23 ± 4.75
50%	81.13 ± 7.15	83.63 ± 9.21	75.10 ± 8.81	85.10 ± 5.73	80.17 ± 8.99

Trust Poisoning

Hostile nodes	AODV	EAURP	DRL-EAURP (paper)	DRL-EAURP (senior)	AR-EAURP
0%	89.17 ± 5.34	87.07 ± 6.41	85.67 ± 8.71	92.20 ± 2.78	89.23 ± 4.73
10%	87.10 ± 7.19	83.53 ± 9.72	78.83 ± 11.29	90.83 ± 4.44	80.47 ± 8.18
20%	84.27 ± 5.43	61.13 ± 14.01	78.47 ± 11.74	89.30 ± 5.79	72.20 ± 17.50
30%	83.67 ± 9.62	56.90 ± 14.72	70.47 ± 10.84	86.40 ± 6.85	54.60 ± 12.97
40%	82.50 ± 7.31	48.13 ± 12.71	57.83 ± 8.58	85.23 ± 7.72	58.00 ± 14.81
50%	81.13 ± 7.15	48.50 ± 11.02	54.00 ± 7.28	85.10 ± 5.73	47.03 ± 14.20

E3 detection and paired delivery analysis

EAURP and the two DRL-EAURP baselines do not use the AR GAN detector; their detector-specific fields are not comparable. AR detector values below are the project's per-window metrics.

Attack	Hostile nodes	AR window TPR	AR window FPR	AR score AUC	AR−EAURP PDR (pp)	Paired 95% CI (pp)	AR wins
blackhole	0%	0.00 ± 0.00	5.51 ± 2.94	0.50 ± 0.00	2.17	[-4.68, 9.02]	4/5
blackhole	10%	96.00 ± 8.94	6.63 ± 3.82	0.98 ± 0.02	-7.67	[-21.75, 6.41]	2/5
blackhole	20%	100.00 ± 0.00	7.26 ± 6.98	0.97 ± 0.03	-0.33	[-9.36, 8.69]	2/5
blackhole	30%	93.42 ± 8.17	10.97 ± 5.12	0.97 ± 0.04	-2.33	[-11.33, 6.66]	2/5
blackhole	40%	100.00 ± 0.00	13.69 ± 13.12	0.97 ± 0.06	3.43	[-9.16, 16.03]	2/5

Attack	Hostile nodes	AR window TPR	AR window FPR	AR score AUC	AR−EAURP PDR (pp)	Paired 95% CI (pp)	AR wins
blackhole	50%	99.05 ± 2.13	12.29 ± 12.89	0.98 ± 0.05	1.30	[-6.35, 8.95]	2/5
grayhole	0%	0.00 ± 0.00	5.51 ± 2.94	0.50 ± 0.00	2.17	[-4.68, 9.02]	4/5
grayhole	10%	6.19 ± 8.52	5.89 ± 3.45	0.55 ± 0.15	-3.10	[-8.64, 2.44]	1/5
grayhole	20%	2.36 ± 3.24	3.26 ± 2.49	0.62 ± 0.11	6.67	[-3.90, 17.24]	4/5
grayhole	30%	7.82 ± 4.98	5.24 ± 5.14	0.55 ± 0.07	1.23	[-9.06, 11.53]	2/5
grayhole	40%	9.83 ± 5.31	4.04 ± 4.53	0.59 ± 0.12	-2.10	[-12.75, 8.55]	2/5
grayhole	50%	9.29 ± 5.79	3.94 ± 3.54	0.56 ± 0.11	-3.47	[-10.35, 3.42]	1/5
trust poisoning	0%	0.00 ± 0.00	5.51 ± 2.94	0.50 ± 0.00	2.17	[-4.68, 9.02]	4/5
trust poisoning	10%	3.33 ± 7.45	7.40 ± 1.32	0.46 ± 0.17	-3.07	[-16.22, 10.09]	3/5
trust poisoning	20%	8.30 ± 6.04	7.64 ± 5.16	0.51 ± 0.03	11.07	[1.35, 20.79]	5/5
trust poisoning	30%	12.31 ± 8.77	5.00 ± 4.06	0.51 ± 0.13	-2.30	[-14.62, 10.02]	2/5
trust poisoning	40%	10.94 ± 3.96	7.70 ± 2.95	0.49 ± 0.04	9.87	[-1.37, 21.10]	4/5
trust poisoning	50%	10.17 ± 3.20	7.21 ± 5.36	0.39 ± 0.05	-1.47	[-21.89, 18.95]	2/5

The paired interval is for AR-EAURP minus EAURP PDR within that attack/prevalence condition. A positive point estimate is not, by itself, evidence of superiority; inspect the interval and the five paired outcomes in the raw check-points.

E4 — Energy mode

Configuration	PDR	Residual energy (J)	Dead nodes
AODV / battery	89.17 ± 5.34	6022.93 ± 2.25	0.00 ± 0.00
AODV / solar	89.17 ± 5.34	7640.87 ± 207.91	0.00 ± 0.00
EAURP / battery	88.83 ± 2.98	6021.79 ± 1.85	0.00 ± 0.00
EAURP / solar	87.07 ± 6.41	7640.65 ± 207.70	0.00 ± 0.00
DRL-EAURP (paper) / battery	82.93 ± 13.70	6021.05 ± 4.12	0.00 ± 0.00
DRL-EAURP (paper) / solar	85.67 ± 8.71	7638.78 ± 208.57	0.00 ± 0.00
DRL-EAURP (senior) / battery	89.10 ± 4.57	6021.75 ± 2.39	0.00 ± 0.00
DRL-EAURP (senior) / solar	92.20 ± 2.78	7639.39 ± 207.71	0.00 ± 0.00
AR-EAURP / battery	87.47 ± 4.41	6022.38 ± 2.18	0.00 ± 0.00
AR-EAURP / solar	89.23 ± 4.73	7639.45 ± 207.81	0.00 ± 0.00

E4b — Network lifetime

Configuration	First death (s)	80% dead (s)	Residual energy (J)
AODV / battery	422.65 ± 0.56	424.55 ± 0.15	0.00 ± 0.00
AODV / solar	432.10 ± 2.28	censored / mixed	1395.71 ± 166.72
EAURP / battery	423.30 ± 0.58	424.90 ± 0.20	0.00 ± 0.00
EAURP / solar	432.84 ± 1.55	censored / mixed	1370.50 ± 190.10

Configuration	First death (s)	80% dead (s)	Residual energy (J)
DRL-EAURP (paper) / battery	423.50 ± 0.38	424.98 ± 0.33	0.00 ± 0.00
DRL-EAURP (paper) / solar	432.20 ± 2.51	censored / mixed	1402.79 ± 212.21
DRL-EAURP (senior) / battery	422.89 ± 0.58	424.76 ± 0.11	0.00 ± 0.00
DRL-EAURP (senior) / solar	432.12 ± 2.34	censored / mixed	1394.65 ± 167.13
AR-EAURP / battery	423.60 ± 0.69	425.12 ± 0.34	0.00 ± 0.00
AR-EAURP / solar	432.79 ± 1.90	censored / mixed	1360.29 ± 212.01

E5 — AR-EAURP component ablations

Configuration	PDR	Window TPR	Window FPR	Score AUC
C full	83.90 ± 8.08	7.82 ± 4.98	5.24 ± 5.14	0.55 ± 0.07
C without GAN	85.63 ± 8.04	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.00
C without LSTM	83.97 ± 7.25	8.89 ± 8.43	5.92 ± 3.67	0.56 ± 0.03
C without CMDP	84.50 ± 7.35	6.77 ± 8.57	5.55 ± 5.21	0.60 ± 0.09
C bare	82.67 ± 2.84	0.00 ± 0.00	0.00 ± 0.00	0.50 ± 0.00
C without energy excuse	83.90 ± 8.08	7.82 ± 4.98	5.24 ± 5.14	0.55 ± 0.07

E5b — CMDP PDR floor

Configuration	PDR	Final CMDP λ	Peak CMDP λ
C floor=0.70	70.77 ± 20.20	0.43 ± 0.54	0.50 ± 0.50
C floor=0.80	88.63 ± 4.07	0.70 ± 0.65	0.86 ± 0.54
C floor=0.90	83.90 ± 8.08	1.69 ± 0.76	1.71 ± 0.72
C floor=0.95	78.20 ± 12.22	2.33 ± 0.99	2.33 ± 0.99

E6 — Security overhead

Configuration	PDR	Crypto overhead (B)	Control bytes	Delay (s)
AODV	83.67 ± 9.62	30856.00 ± 1620.25	180906.20 ± 13723.64	0.21 ± 0.14
EAURP	82.67 ± 2.84	30654.40 ± 494.74	224338.60 ± 10550.78	0.27 ± 0.08
DRL-EAURP (paper)	82.60 ± 10.65	30699.20 ± 1761.51	223893.80 ± 31384.10	0.23 ± 0.10
DRL-EAURP (senior)	86.40 ± 6.85	31326.40 ± 1143.28	206686.40 ± 14286.77	0.21 ± 0.06
AR-EAURP	83.90 ± 8.08	30906.40 ± 1355.82	218395.00 ± 14430.38	0.28 ± 0.14
C without ECC	84.03 ± 4.51	0.00 ± 0.00	230747.60 ± 24773.70	0.24 ± 0.16

Interpretation and limits

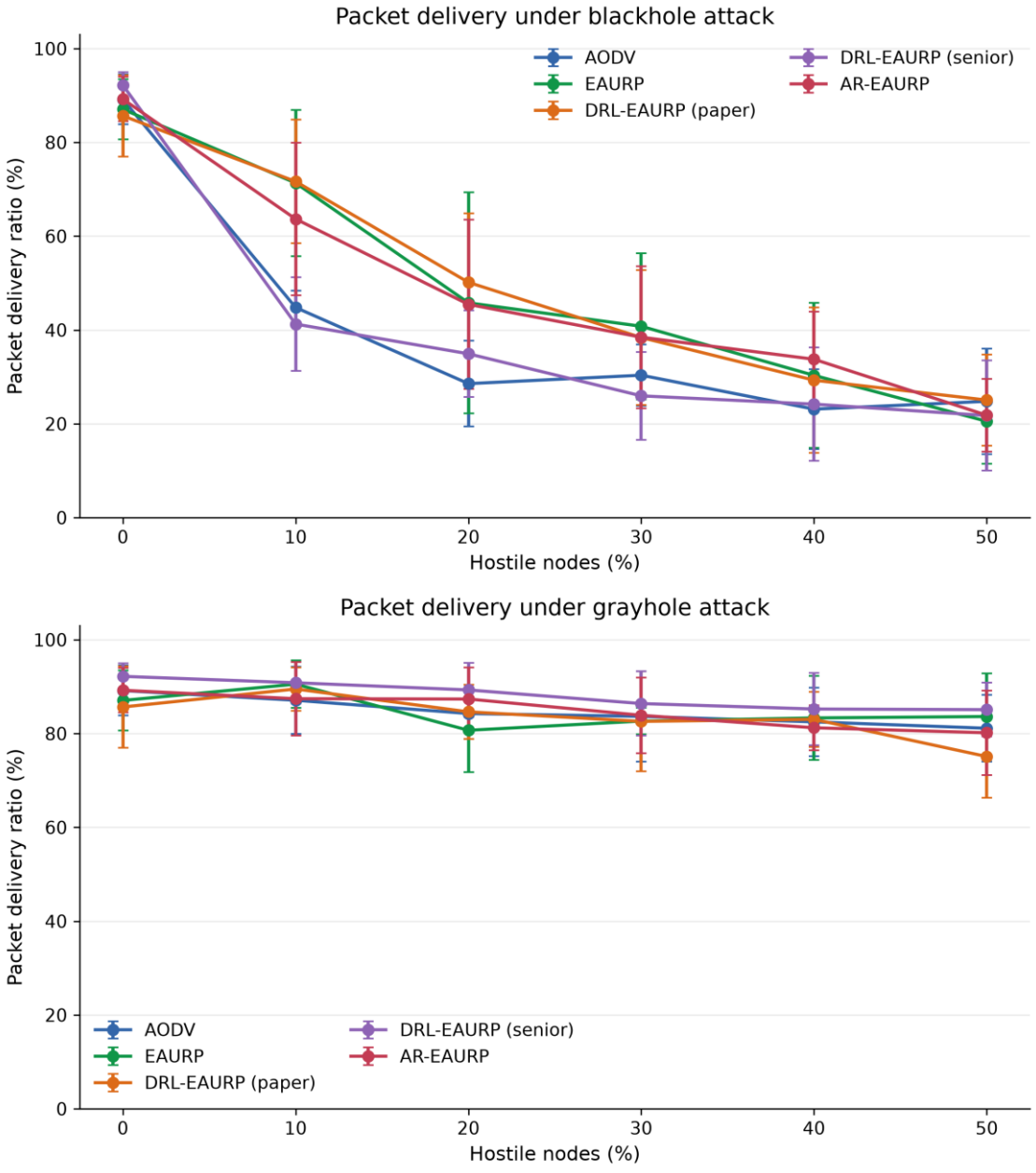
1. E3 is the direct attack-prevalence comparison. Read the protocol rows at each attack and prevalence rather than collapsing all hostile fractions into one score.
2. PDR is application packets received divided by packets sent in the measured window, with arrivals through cooldown. Delay and throughput are also application-level metrics.
3. The planned design uses five run numbers per condition. The completion-status table reports the actual number used for each experiment; single-run conditions have no empirical SD or confidence interval and support

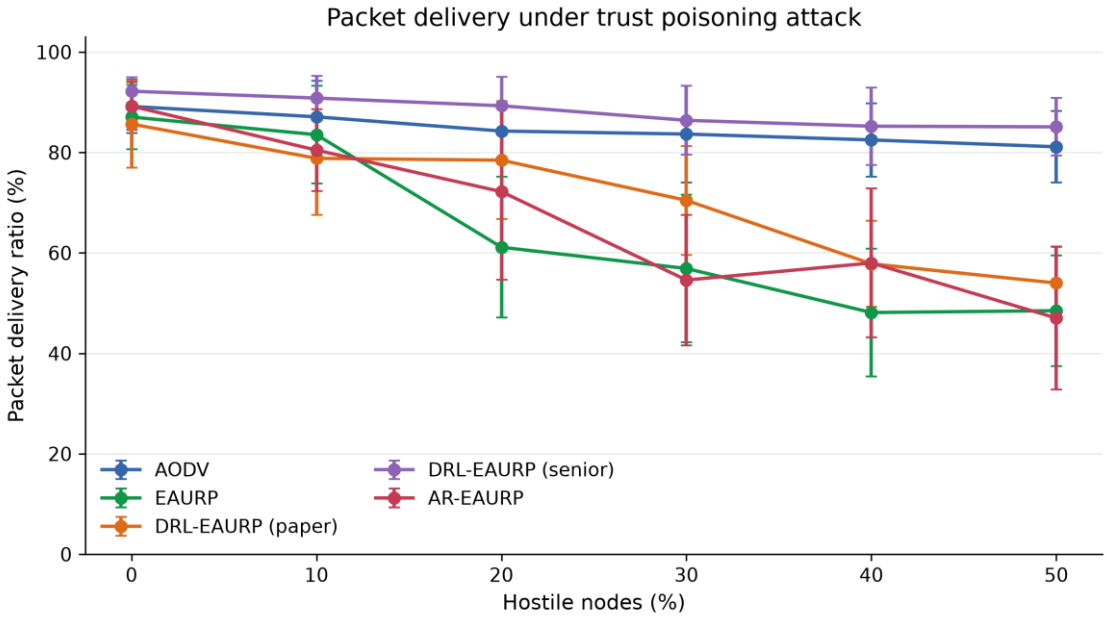
descriptive observations only. Multiple comparisons across many cells are exploratory and no family-wise error correction is applied.

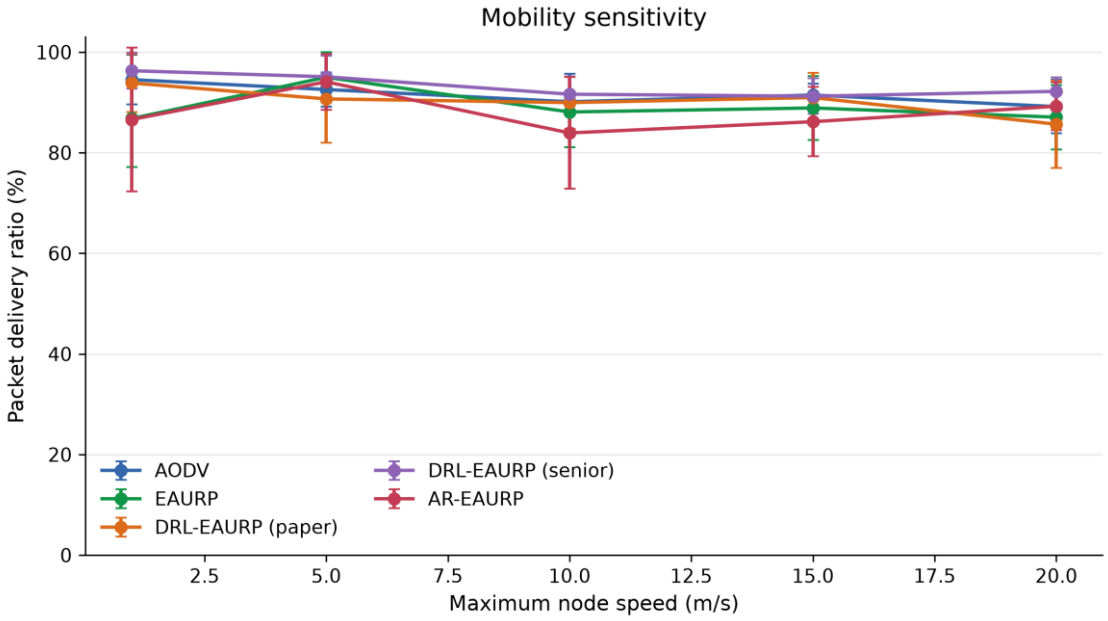
4. Pairing matches the configured ns-3 seed and run number across protocols. It does not force event-by-event identical random draws; protocol-specific event paths can consume random streams differently.
5. AR's TPR/FPR/AUC are detector measurements. Baseline protocols do not implement that detector, so a zero/default baseline detector value must not be interpreted as perfect classification or compared as a detector score.
6. The lifetime experiment includes censored runs. Censored observations must not be treated as observed death times; the raw observed/censored flags are preserved in the CSV.
7. These simulations support claims only for the stated ns-3 models and factor ranges. They do not establish real-world wireless performance without calibration and independent replication.

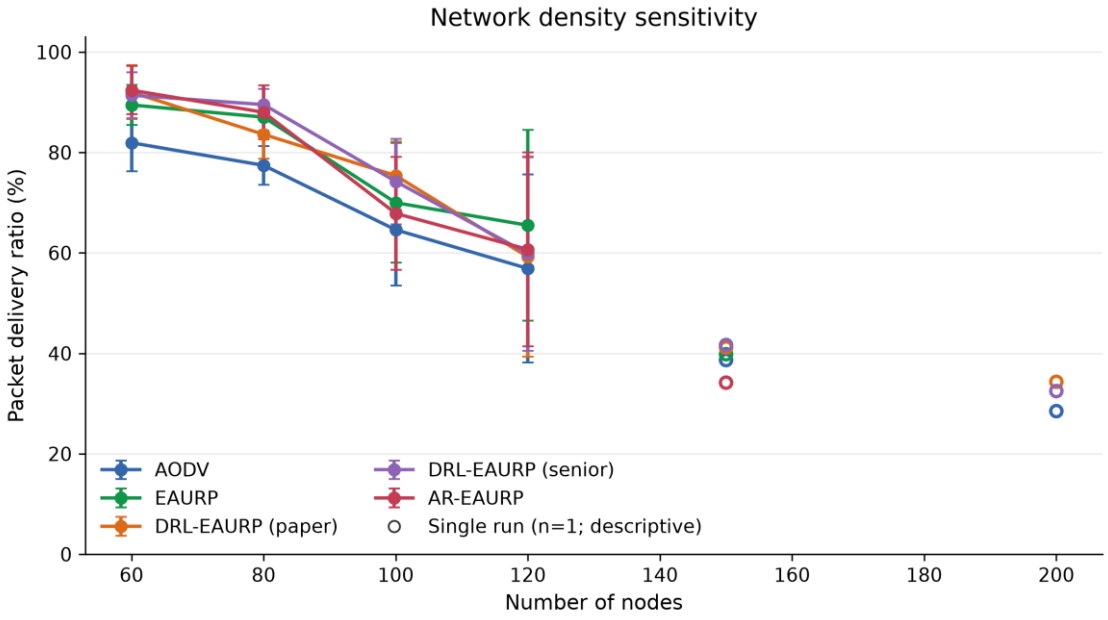
Visual comparisons

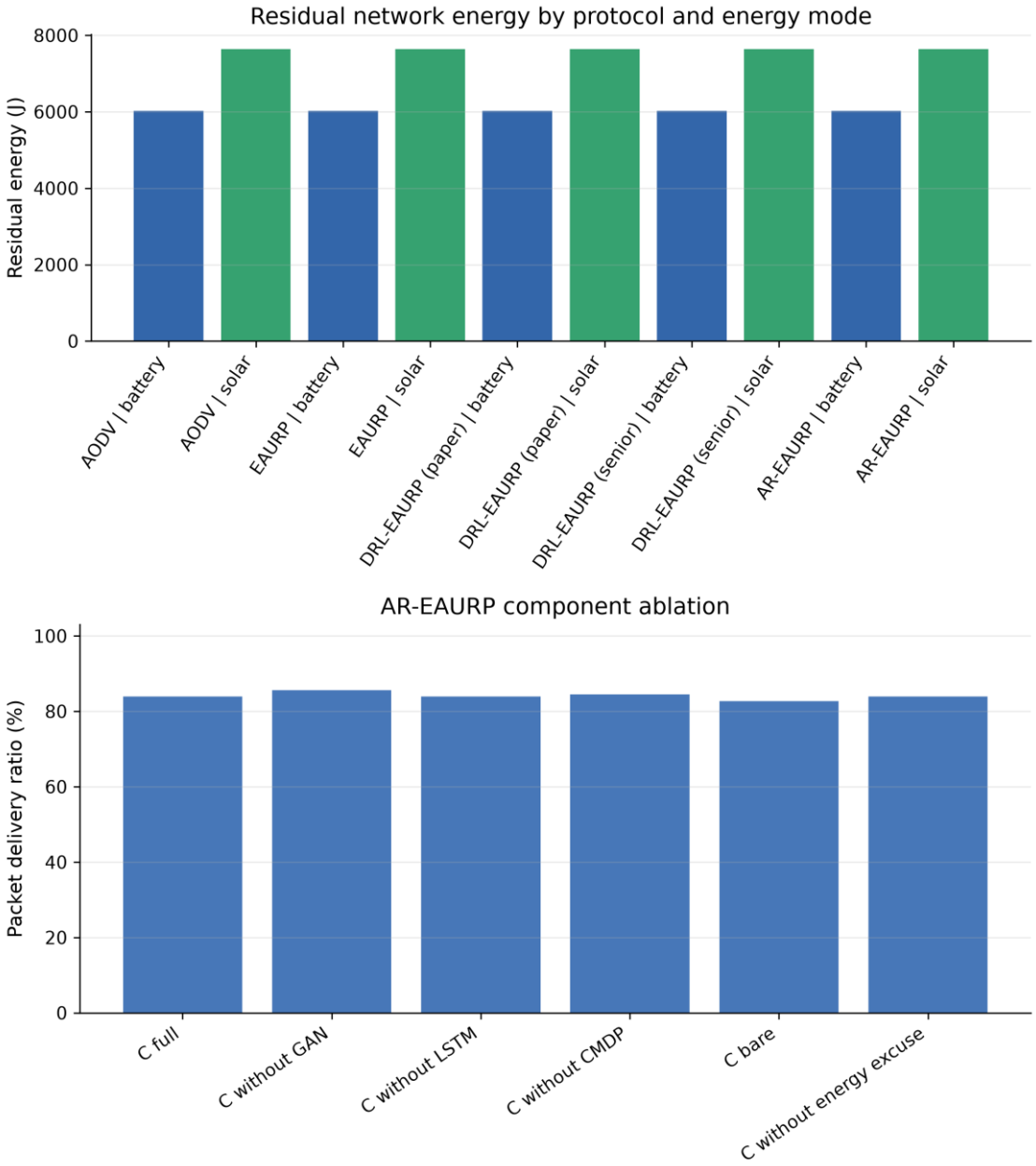
Error bars show sample standard deviation. In the density figure, open circles mark single-run conditions (n=1), which are descriptive only.




Packet delivery under trust poisoning attack
Packet delivery ratio (%)
Hostile nodes (%)
AODV
EAURP
DRL-EAURP (paper)
DRL-EAURP (senior)
AR-EAURP


Mobility sensitivity
Packet delivery ratio (%)
Maximum node speed (m/s)
AODV
EAURP
DRL-EAURP (paper)
DRL-EAURP (senior)
AR-EAURP


Network density sensitivity
Packet delivery ratio (%)
Number of nodes
AODV
EAURP
DRL-EAURP (paper)
DRL-EAURP (senior)
AR-EAURP
Single run (n=1; descriptive)



Files and provenance

- Completed 179 of 181 planned conditions and 863 of 905 planned ns-3 runs; per-experiment coverage is tabulated above.
- `study_settings.json`: complete configuration and factor-level manifest.
- `configuration_settings.csv`: full settings plus aggregate result for every condition.
- `run_settings.csv`: settings and raw checkpoint provenance for each completed individual run.
- `pairwise_e3_comparisons.csv`: all ten matched protocol-pair PDR contrasts across the 18 attack/prevalence conditions, with 95% intervals and per-run differences.
- `csv/`: aggregate condition-level results, including simulator-reported cfg_* settings and sample SDs.
- `raw/`: checksummed per-run metric checkpoints used for the paired E3 PDR analysis.
- `report.md`, `report.html`, and `figures/`: native experiment-runner tables, plot scripts, and rendered SVG/PNG figures.
- `paper_results.md`: this expanded paper-style summary.

Separate CMDP commissioning study

Question

Does training the CMDP route selector on clean commissioning traffic improve AR-EAURP delivery under a grayhole attack, compared with EAURP?

Experiment

- ns-3.45 native C++ simulation
- 24 nodes, 6 CBR flows, 25% grayhole nodes, 420 m square area
- 120 s simulation, 20 s warm-up, fixed seed 1
- Matched run numbers 21–50; each protocol uses the same run number
- AR-EAURP trains its detector/forecaster for 240 s on a separate clean commissioning network before evaluation
- The CMDP is now also trained on route outcomes during that clean phase
- EAURP runs without the AR models, as the baseline

Result

Measure	EAURP	AR-EAURP, CMDP trained in commissioning
Mean PDR	79.81%	80.97%
Mean delay	0.256 s	0.288 s
Mean route discovery failures	11.3	10.9
AR detector window TPR	—	4.69%
AR detector window FPR	—	4.94%
AR detector window AUC	—	0.586

The matched mean PDR difference is **+1.16 percentage points** for AR-EAURP. AR-EAURP had higher PDR in 16 of 30 pairs. The paired 95% confidence interval is **−2.97 to +5.30 percentage points**, so the experiment does not establish that the true PDR difference is positive. Mean delay was 32 ms higher for AR-EAURP (paired 95% CI **−21 to +84 ms**). Treat both as inconclusive results, not proof of superiority.

EAURP has no comparable AR anomaly detector in this experiment. Its zero TPR and FPR fields and AUC 0.5 are output defaults, not detector measurements.

Files

- `cmdp_commissioning_holdout/`: matched runs 21–30, JSON and full logs
- `cmdp_commissioning_extended/`: matched runs 31–50, JSON and full logs
- `tuning/`: pre-fix exploratory runs 6–10, including the no-CMDP ablation
- `exploration_tuning/`: pre-fix epsilon-greedy and greedy CMDP ablations
- `holdout/`: pre-fix validation runs 11–20, including the greedy CMDP ablation

The older tuning and validation folders tested the original evaluation-only CMDP. The commissioning-trained implementation was introduced afterward and was evaluated on the separate run range 21–50 above. The current scripts run the commissioning-trained implementation.

Each JSON file contains the scenario configuration and raw summary metrics; the adjacent `.log` contains the native simulator output.

The earlier greedy CMDP candidate did not outperform EAURP on its 10 held-out pairs (−2.80 PDR points, 4/10 wins). Disabling CMDP was also worse on those pairs. Lowering exploration was worse on the tuning runs. These ablations are why commissioning training was tested instead of selecting an epsilon setting that only looked favorable on a small seed subset.

Reproduction

From the project root in WSL, set `NS3_HOME` to the ns-3.45 checkout used for this build, then run:

```
export NS3_HOME="$HOME/.cache/ar-eaurp/ns-3.45"
bash ns3/scripts/run_cmdp_commissioning_study.sh \
  results/optimization_study/reproduction 21 50
```

Build and verify with:

```
bash ns3/scripts/setup.sh
bash ns3/scripts/run.sh --verify=1
```

The native verification suite passed after the CMDP commissioning-training change.

All saved exploratory optimization groups

The following table inventories every saved JSON group in the optimization study. These exploratory groups include earlier code states and different run ranges, and must not be pooled into a single confirmatory comparison. The held-out commissioning estimate is the separately reported 30-pair result above.

Folder / candidate	Saved runs	PDR mean ± SD (%)	Delay mean ± SD (s)
cmdp commissioning extended / ar-commission-trained	20	80.98 ± 8.48	0.290 ± 0.163
cmdp commissioning extended / eaurp	20	80.44 ± 6.70	0.252 ± 0.076
cmdp commissioning holdout / ar-commission-trained	10	80.93 ± 6.86	0.282 ± 0.138
cmdp commissioning holdout / eaurp	10	78.53 ± 10.72	0.264 ± 0.089
exploration tuning / ar-greedy	5	83.70 ± 10.06	0.255 ± 0.125
exploration tuning / ar-low-exploration	5	79.67 ± 9.66	0.287 ± 0.149
holdout / ar-default	10	80.65 ± 9.59	0.280 ± 0.060
holdout / ar-greedy	10	81.75 ± 8.80	0.283 ± 0.113
holdout / ar-no-cmdp	10	77.82 ± 10.25	0.313 ± 0.144
holdout / eaurp	10	84.55 ± 5.82	0.286 ± 0.112
tuning / ar-default	5	85.00 ± 5.34	0.234 ± 0.116
tuning / ar-no-cmdp	5	86.53 ± 11.30	0.286 ± 0.161
tuning / eaurp	5	81.13 ± 11.64	0.287 ± 0.131

Earlier native five-run validation

This earlier snapshot used seed 1 and runs 1–5, 24 nodes, six flows, 420 m square, 25% gray-hole prevalence, a 120 s scenario duration with 20 s excluded warm-up (100 s measured), and 240 s AR commissioning. It is separate from the later E1–E6 review matrix and the 30-pair commissioning follow-up.

Metric	EAURP	AR-EAURP
PDR, mean ± sample SD	83.73% ± 4.23%	81.83% ± 4.60%
Delay	257 ± 86 ms	292 ± 108 ms
Detector TPR	Not applicable	6.79% ± 3.32%
Detector FPR	Not applicable	5.83% ± 6.14%
Detector score AUC	Not applicable	0.55 ± 0.08
Two-witness contested TPR	Not applicable	0% in all five runs

The paired PDR difference was −1.90 percentage points, 95% CI −8.46 to +4.66: inconclusive. The LSTM lost to persistence in all five commissioning evaluations and was not selected for routing. Source: `results/final_review_ns3_results/FINAL_NS3_EVALUATION_REPORT.md` and its `native_long_runs` records.

Historical Python / prior Colab-review result

These saved Python full-profile results concern the earlier simulator and workload. They are historical context and are not pooled with the current ns-3 results. The table lists the 30% gray-hole slice across five saved runs. Values are mean ± sample SD. The execution location is not established by a CSV; this is the Python result used for the prior Colab-review context.

Protocol	PDR (%)	Detection TPR (%)	Detection FPR (%)
AODV (reference baseline)	46.47 ± 5.75	0.00 ± 0.00	0.00 ± 0.00
EAURP (base substrate)	47.92 ± 5.24	0.00 ± 0.00	0.00 ± 0.00
A: DRL-EAURP (paper)	45.96 ± 6.01	0.00 ± 0.00	0.00 ± 0.00
B: DRL-EAURP (senior code)	46.41 ± 5.93	0.00 ± 0.00	0.00 ± 0.00
C: AR-EAURP (advancement)	50.25 ± 5.86	29.33 ± 7.42	1.43 ± 0.90

AR-EAURP's historical PDR point estimate was 2.32 percentage points above EAURP. That observation motivated the native evaluation, but does not establish that the native implementation retains the benefit: implementations, radio/channel models, topology, traffic and detector observation processes differ.

Automated verification and data audit

The native integration verification workflow had passed during the implementation work. The checks cover repeatability of simulated metrics; protected UDP delivery and authentication; delivery for the protocol variants; Q-learning updates; disconnected-flow feedback during cooldown; trained model/CMDP activity; energy shutdown/recharge; black-hole/gray-hole behavior; sample-standard-deviation arithmetic; invalid configuration rejection; matching checkpoint reuse; corrupted checkpoint recomputation; changed source/configuration identity rejection; CSV escaping; content hashing; and the E1–E6 smoke configuration inventory. These engineering checks verify implementation properties and do not establish comparative protocol superiority. They were not rerun during this document-formatting task.

The separate native unit-test source inventories the following cases. This inventory is not a claim of a fresh unit-suite execution during this editing session:

- control headers serialise and deserialise.
- the dense network learns XOR.
- the LSTM learns a periodic signal.
- a gray-hole evades the scalar trust threshold.
- forecast validation requires disjoint temporal windows per node.
- X25519 peers agree and authenticated payloads reject tampering.
- tabular feedback changes routes and endpoint trust uses endpoint counters.
- warmup packets do not inflate evaluation PDR.

The earlier report audit reconciled all 863 ledger-referenced checkpoint hashes, aggregate means/sample SDs and all 180 E3 paired differences. The condition and per-run ledgers preserve exact settings. The main report is a composite of verified checkpoints; the root manifest's original `e3,e4,e5,e6` selection does not exclude the separately aggregated E1 and partial E2 data.

All 180 matched E3 protocol contrasts

Each row is PDR(A) minus PDR(B) in percentage points, using five paired run numbers. Intervals are two-sided Student t 95% intervals and are unadjusted across this exploratory family. Wins count A above B; ties are retained. Zero-hostile rows appearing under multiple attack labels share an underlying clean condition and must not be interpreted as separate independent confirmation.

For compact tables, DRL-paper denotes DRL-EAURP (paper), and DRL-senior denotes DRL-EAURP (senior).

blackhole — 0% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	+2.10	[-2.72, +6.92]	4	0	5
AODV − DRL-paper	+3.50	[-8.01, +15.01]	3	0	5
AODV − DRL-senior	-3.03	[-6.74, +0.67]	0	0	5
AODV − AR-EAURP	-0.07	[-3.12, +2.99]	3	0	5
EAURP − DRL-paper	+1.40	[-13.14, +15.94]	3	0	5
EAURP − DRL-senior	-5.13	[-11.30, +1.04]	0	0	5
EAURP − AR-EAURP	-2.17	[-9.02, +4.68]	1	0	5
DRL-paper − DRL-senior	-6.53	[-17.11, +4.04]	1	0	5
DRL-paper − AR-EAURP	-3.57	[-15.67, +8.54]	1	0	5
DRL-senior − AR-EAURP	+2.97	[-0.78, +6.71]	4	0	5

blackhole — 10% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	-26.53	[-47.64, -5.42]	1	0	5
AODV − DRL-paper	-26.87	[-45.40, -8.34]	0	0	5
AODV − DRL-senior	+3.53	[-5.62, +12.68]	3	0	5
AODV − AR-EAURP	-18.87	[-42.75, +5.02]	1	0	5
EAURP − DRL-paper	-0.33	[-7.07, +6.41]	3	0	5
EAURP − DRL-senior	+30.07	[+5.84, +54.29]	5	0	5
EAURP − AR-EAURP	+7.67	[-6.41, +21.75]	3	0	5
DRL-paper − DRL-senior	+30.40	[+9.55, +51.25]	5	0	5
DRL-paper − AR-EAURP	+8.00	[-2.93, +18.93]	4	0	5
DRL-senior − AR-EAURP	-22.40	[-50.16, +5.36]	0	0	5

blackhole — 20% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	-17.23	[-50.36, +15.90]	1	0	5
AODV − DRL-paper	-21.60	[-45.06, +1.86]	1	0	5
AODV − DRL-senior	-6.37	[-8.40, -4.33]	0	0	5
AODV − AR-EAURP	-16.90	[-43.13, +9.33]	1	0	5
EAURP − DRL-paper	-4.37	[-18.38, +9.65]	3	0	5
EAURP − DRL-senior	+10.87	[-20.58, +42.32]	4	0	5
EAURP − AR-EAURP	+0.33	[-8.69, +9.36]	3	0	5
DRL-paper − DRL-senior	+15.23	[-6.98, +37.44]	4	0	5
DRL-paper − AR-EAURP	+4.70	[-1.83, +11.23]	4	0	5
DRL-senior − AR-EAURP	-10.53	[-35.15, +14.09]	1	0	5

blackhole — 30% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	-10.43	[-28.63, +7.77]	1	0	5
AODV − DRL-paper	-8.03	[-25.89, +9.82]	1	0	5
AODV − DRL-senior	+4.40	[-5.57, +14.37]	3	0	5
AODV − AR-EAURP	-8.10	[-25.26, +9.06]	1	0	5
EAURP − DRL-paper	+2.40	[-4.29, +9.09]	4	0	5
EAURP − DRL-senior	+14.83	[-11.77, +41.44]	4	0	5
EAURP − AR-EAURP	+2.33	[-6.66, +11.33]	3	0	5
DRL-paper − DRL-senior	+12.43	[-11.83, +36.70]	4	0	5
DRL-paper − AR-EAURP	-0.07	[-4.68, +4.55]	2	2	5
DRL-senior − AR-EAURP	-12.50	[-35.45, +10.45]	1	0	5

blackhole — 40% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	-7.20	[-21.37, +6.97]	1	0	5
AODV − DRL-paper	-6.17	[-21.99, +9.65]	1	0	5
AODV − DRL-senior	-1.07	[-8.06, +5.93]	2	0	5
AODV − AR-EAURP	-10.63	[-18.34, -2.93]	0	0	5
EAURP − DRL-paper	+1.03	[-6.98, +9.05]	3	0	5
EAURP − DRL-senior	+6.13	[-7.80, +20.07]	4	0	5
EAURP − AR-EAURP	-3.43	[-16.03, +9.16]	3	0	5
DRL-paper − DRL-senior	+5.10	[-12.96, +23.16]	3	0	5
DRL-paper − AR-EAURP	-4.47	[-15.80, +6.86]	1	0	5
DRL-senior − AR-EAURP	-9.57	[-23.12, +3.99]	0	0	5

blackhole — 50% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	+4.27	[-7.64, +16.17]	3	0	5
AODV − DRL-paper	-0.27	[-16.46, +15.92]	3	0	5
AODV − DRL-senior	+3.03	[-6.16, +12.23]	3	0	5
AODV − AR-EAURP	+2.97	[-8.23, +14.16]	4	0	5
EAURP − DRL-paper	-4.53	[-9.38, +0.32]	1	0	5
EAURP − DRL-senior	-1.23	[-11.16, +8.69]	1	0	5
EAURP − AR-EAURP	-1.30	[-8.95, +6.35]	3	0	5
DRL-paper − DRL-senior	+3.30	[-8.93, +15.53]	3	0	5
DRL-paper − AR-EAURP	+3.23	[-7.94, +14.41]	4	0	5
DRL-senior − AR-EAURP	-0.07	[-14.64, +14.51]	3	0	5

grayhole — 0% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	+2.10	[-2.72, +6.92]	4	0	5
AODV − DRL-paper	+3.50	[-8.01, +15.01]	3	0	5
AODV − DRL-senior	-3.03	[-6.74, +0.67]	0	0	5
AODV − AR-EAURP	-0.07	[-3.12, +2.99]	3	0	5
EAURP − DRL-paper	+1.40	[-13.14, +15.94]	3	0	5
EAURP − DRL-senior	-5.13	[-11.30, +1.04]	0	0	5
EAURP − AR-EAURP	-2.17	[-9.02, +4.68]	1	0	5
DRL-paper − DRL-senior	-6.53	[-17.11, +4.04]	1	0	5
DRL-paper − AR-EAURP	-3.57	[-15.67, +8.54]	1	0	5
DRL-senior − AR-EAURP	+2.97	[-0.78, +6.71]	4	0	5

grayhole — 10% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	-3.43	[-6.99, +0.13]	0	0	5
AODV − DRL-paper	-2.40	[-8.19, +3.39]	2	0	5
AODV − DRL-senior	-3.73	[-8.61, +1.14]	1	0	5
AODV − AR-EAURP	-0.33	[-4.73, +4.07]	2	0	5
EAURP − DRL-paper	+1.03	[-2.07, +4.13]	3	0	5
EAURP − DRL-senior	-0.30	[-4.15, +3.55]	2	0	5
EAURP − AR-EAURP	+3.10	[-2.44, +8.64]	3	1	5
DRL-paper − DRL-senior	-1.33	[-6.43, +3.76]	2	0	5
DRL-paper − AR-EAURP	+2.07	[-3.75, +7.88]	3	0	5
DRL-senior − AR-EAURP	+3.40	[-4.29, +11.09]	3	0	5

grayhole — 20% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	+3.57	[-5.73, +12.86]	3	0	5
AODV − DRL-paper	-0.37	[-10.79, +10.06]	2	0	5
AODV − DRL-senior	-5.03	[-10.11, +0.04]	1	0	5
AODV − AR-EAURP	-3.10	[-7.01, +0.81]	1	0	5
EAURP − DRL-paper	-3.93	[-20.25, +12.39]	1	0	5
EAURP − DRL-senior	-8.60	[-21.21, +4.01]	2	0	5
EAURP − AR-EAURP	-6.67	[-17.24, +3.90]	1	0	5
DRL-paper − DRL-senior	-4.67	[-15.99, +6.66]	1	0	5
DRL-paper − AR-EAURP	-2.73	[-13.41, +7.94]	2	0	5
DRL-senior − AR-EAURP	+1.93	[-4.17, +8.04]	3	0	5

grayhole — 30% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	+1.00	[-9.59, +11.59]	3	0	5
AODV − DRL-paper	+1.07	[-17.17, +19.30]	1	0	5
AODV − DRL-senior	-2.73	[-8.01, +2.54]	1	0	5
AODV − AR-EAURP	-0.23	[-8.35, +7.89]	2	0	5
EAURP − DRL-paper	+0.07	[-16.08, +16.21]	2	0	5
EAURP − DRL-senior	-3.73	[-12.01, +4.55]	1	0	5
EAURP − AR-EAURP	-1.23	[-11.53, +9.06]	3	0	5
DRL-paper − DRL-senior	-3.80	[-17.57, +9.97]	3	0	5
DRL-paper − AR-EAURP	-1.30	[-13.37, +10.77]	3	0	5
DRL-senior − AR-EAURP	+2.50	[-3.66, +8.66]	3	0	5

grayhole — 40% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	-0.83	[-5.62, +3.95]	2	0	5
AODV − DRL-paper	-0.53	[-13.73, +12.66]	2	0	5
AODV − DRL-senior	-2.73	[-6.67, +1.21]	1	0	5
AODV − AR-EAURP	+1.27	[-7.54, +10.08]	2	0	5
EAURP − DRL-paper	+0.30	[-14.70, +15.30]	2	0	5
EAURP − DRL-senior	-1.90	[-8.28, +4.48]	2	0	5
EAURP − AR-EAURP	+2.10	[-8.55, +12.75]	3	0	5
DRL-paper − DRL-senior	-2.20	[-14.40, +10.00]	3	0	5
DRL-paper − AR-EAURP	+1.80	[-3.21, +6.81]	3	0	5
DRL-senior − AR-EAURP	+4.00	[-4.63, +12.63]	3	0	5

grayhole — 50% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	-2.50	[-5.75, +0.75]	1	0	5
AODV − DRL-paper	+6.03	[+0.44, +11.63]	5	0	5
AODV − DRL-senior	-3.97	[-7.86, -0.08]	1	0	5
AODV − AR-EAURP	+0.97	[-4.49, +6.43]	3	0	5
EAURP − DRL-paper	+8.53	[+2.98, +14.09]	5	0	5
EAURP − DRL-senior	-1.47	[-7.62, +4.68]	1	0	5
EAURP − AR-EAURP	+3.47	[-3.42, +10.35]	4	0	5
DRL-paper − DRL-senior	-10.00	[-18.82, -1.18]	0	0	5
DRL-paper − AR-EAURP	-5.07	[-15.55, +5.42]	1	1	5
DRL-senior − AR-EAURP	+4.93	[+0.48, +9.39]	5	0	5

trust poisoning — 0% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	+2.10	[-2.72, +6.92]	4	0	5
AODV − DRL-paper	+3.50	[-8.01, +15.01]	3	0	5
AODV − DRL-senior	-3.03	[-6.74, +0.67]	0	0	5
AODV − AR-EAURP	-0.07	[-3.12, +2.99]	3	0	5
EAURP − DRL-paper	+1.40	[-13.14, +15.94]	3	0	5
EAURP − DRL-senior	-5.13	[-11.30, +1.04]	0	0	5
EAURP − AR-EAURP	-2.17	[-9.02, +4.68]	1	0	5
DRL-paper − DRL-senior	-6.53	[-17.11, +4.04]	1	0	5
DRL-paper − AR-EAURP	-3.57	[-15.67, +8.54]	1	0	5
DRL-senior − AR-EAURP	+2.97	[-0.78, +6.71]	4	0	5

trust poisoning — 10% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	+3.57	[-8.87, +16.00]	2	0	5
AODV − DRL-paper	+8.27	[-8.85, +25.38]	4	0	5
AODV − DRL-senior	-3.73	[-8.61, +1.14]	1	0	5
AODV − AR-EAURP	+6.63	[-1.56, +14.82]	4	0	5
EAURP − DRL-paper	+4.70	[-6.41, +15.81]	3	0	5
EAURP − DRL-senior	-7.30	[-17.25, +2.65]	0	0	5
EAURP − AR-EAURP	+3.07	[-10.09, +16.22]	2	0	5
DRL-paper − DRL-senior	-12.00	[-26.92, +2.92]	0	0	5
DRL-paper − AR-EAURP	-1.63	[-21.59, +18.32]	3	0	5
DRL-senior − AR-EAURP	+10.37	[+1.70, +19.03]	4	0	5

trust poisoning — 20% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	+23.13	[+1.80, +44.46]	5	0	5
AODV − DRL-paper	+5.80	[-8.98, +20.58]	3	0	5
AODV − DRL-senior	-5.03	[-10.11, +0.04]	1	0	5
AODV − AR-EAURP	+12.07	[-11.35, +35.48]	3	0	5
EAURP − DRL-paper	-17.33	[-34.79, +0.12]	1	0	5
EAURP − DRL-senior	-28.17	[-50.05, -6.28]	0	0	5
EAURP − AR-EAURP	-11.07	[-20.79, -1.35]	0	0	5
DRL-paper − DRL-senior	-10.83	[-26.86, +5.19]	1	0	5
DRL-paper − AR-EAURP	+6.27	[-11.56, +24.09]	4	0	5
DRL-senior − AR-EAURP	+17.10	[-6.36, +40.56]	5	0	5

trust poisoning — 30% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	+26.77	[+3.94, +49.59]	5	0	5
AODV − DRL-paper	+13.20	[+3.61, +22.79]	5	0	5
AODV − DRL-senior	-2.73	[-8.01, +2.54]	1	0	5
AODV − AR-EAURP	+29.07	[+9.83, +48.30]	5	0	5
EAURP − DRL-paper	-13.57	[-39.00, +11.86]	2	0	5
EAURP − DRL-senior	-29.50	[-50.27, -8.73]	0	0	5
EAURP − AR-EAURP	+2.30	[-10.02, +14.62]	3	0	5
DRL-paper − DRL-senior	-15.93	[-23.89, -7.97]	0	0	5
DRL-paper − AR-EAURP	+15.87	[-3.39, +35.12]	4	0	5
DRL-senior − AR-EAURP	+31.80	[+13.80, +49.80]	5	0	5

trust poisoning — 40% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	+34.37	[+15.60, +53.13]	5	0	5
AODV − DRL-paper	+24.67	[+6.87, +42.46]	5	0	5
AODV − DRL-senior	-2.73	[-6.67, +1.21]	1	0	5
AODV − AR-EAURP	+24.50	[+1.60, +47.40]	5	0	5
EAURP − DRL-paper	-9.70	[-24.89, +5.49]	2	0	5
EAURP − DRL-senior	-37.10	[-55.07, -19.13]	0	0	5
EAURP − AR-EAURP	-9.87	[-21.10, +1.37]	1	0	5
DRL-paper − DRL-senior	-27.40	[-43.58, -11.22]	0	0	5
DRL-paper − AR-EAURP	-0.17	[-14.74, +14.40]	4	0	5
DRL-senior − AR-EAURP	+27.23	[+6.21, +48.26]	5	0	5

trust poisoning — 50% hostile

A minus B	Mean Δ (pp)	95% CI (pp)	A wins	Ties	n
AODV − EAURP	+32.63	[+16.26, +49.01]	5	0	5
AODV − DRL-paper	+27.13	[+14.14, +40.12]	5	0	5
AODV − DRL-senior	-3.97	[-7.86, -0.08]	1	0	5
AODV − AR-EAURP	+34.10	[+18.00, +50.20]	5	0	5
EAURP − DRL-paper	-5.50	[-21.45, +10.45]	2	0	5
EAURP − DRL-senior	-36.60	[-53.91, -19.29]	0	0	5
EAURP − AR-EAURP	+1.47	[-18.95, +21.89]	3	0	5
DRL-paper − DRL-senior	-31.10	[-42.08, -20.12]	0	0	5
DRL-paper − AR-EAURP	+6.97	[-3.77, +17.71]	4	0	5
DRL-senior − AR-EAURP	+38.07	[+22.15, +53.98]	5	0	5

Consolidated findings and remaining work

1. The native attack matrix and 30-pair follow-up do not establish a reliable overall PDR advantage for AR-EAURP over EAURP.
2. Native gray-hole detection is weak in the selected workload; per-window TPR/FPR/AUC differ from consensus and revocation outcomes.
3. One of 18 AR-EAURP–EAURP attack/prevalence intervals excludes zero before multiplicity correction; it is exploratory evidence requiring confirmation.

4. E2 density is incomplete. The 42 deferred runs are absent; high-density n=1 points do not provide empirical SDs or confidence intervals.
5. Harvesting increases modeled residual energy and changes lifetime endpoints, some of which are censored. It does not directly validate field battery life.
6. Component ablations and CMDP-floor tests show trade-offs, rather than a uniform benefit from every added component.
7. Earlier Python results and native ns-3 outcomes are distinct experiments. Their raw PDR values should not be treated as a direct controlled cross-simulator treatment effect.
8. Next priorities are completing E2, improving detector validation, testing simpler baselines, defining a feasible delivery constraint, and running a confirmatory study with adequate power and planned multiplicity control.

Experiment-by-experiment findings and qualified conclusions

E1: mobility

The five speed levels do not produce a monotonic PDR curve. AR peaks at 94.10% at 5 m/s but is lower than EAURP at 1, 5, 10, and 15 m/s in the descriptive means. At 20 m/s AR is 89.23% versus EAURP 87.07%, while DRL-S is 92.20%. This does not support a universal AR mobility advantage. Between-run variation is large enough that mean rankings should not be treated as deterministic. Delay must be considered alongside delivery because it excludes lost packets.

E2: density and incomplete high-density evidence

AR exceeds EAURP descriptively at 60 and 80 nodes, then falls below it at 100 and 120. At 120 nodes the SD is around nineteen percentage points for both. The n=1 high-density cells cannot support a stability or superiority claim. Density is varied at fixed area and six-flow workload, so neighbor count, contention, control traffic, connectivity, and traffic-per-node effects change together. The current data do not isolate a single cause for degradation. Forty-two missing runs are a coverage limitation rather than negative results.

E3: attacks and policy dependence

Black holes reduce delivery sharply for every policy. At 30% hostile nodes, AR's 38.47% PDR is below EAURP's 40.80%, despite AR window TPR 93.42%. Local detection is therefore not equivalent to successful end-to-end mitigation. Gray holes are less destructive because selective packet eligibility and a 0.25 drop probability affect only part of relayed traffic. At 30%, AR's +1.23-point matched difference has CI [−9.06,+11.53], and DRL-S has the highest descriptive PDR among the five variants. Trust poisoning is particularly harmful to report-sensitive policies. The AODV-style and DRL-S aggregate PDR values match their gray-hole values in this study; this is an outcome under the implemented report-use semantics, not generic immunity to reputation attacks.

Only the 20% trust-poisoning AR–EAURP interval excludes zero before adjustment: +11.07 points, CI [+1.35,+20.79], five wins in five pairs. This is exploratory evidence within a multiple-comparison family. The negative 10% black-hole estimate (−7.67 points) and negative estimates at several gray-hole percentages must be retained. The study cannot be summarized as a consistent AR improvement.

Detection layers and forecast quality

For 30% gray holes, AR window TPR/FPR/AUC are 7.82%/5.24%/0.55. Corroborated contested TPR is 0%, contested FPR 1.18%, revocation TPR 14.29%, and revocation FPR 11.76% as run means. Different state definitions, denominators, and ordinary trust exclusions explain why these are not interchangeable detector accuracy statistics. The same condition has LSTM MAE 0.495 ± 0.409 W, persistence 0.057 ± 0.013 W, and window-mean 0.095 ± 0.028 W. Persistence was selected in all five runs. An LSTM benefit must not be inferred from a fallback-equipped implementation in which the neural predictor lost validation.

E4/E4b: modeled energy and censored lifetime

AR total residual energy rises from 6,022.38 J to 7,639.45 J with harvesting in the short experiment. Added environmental energy is a direct input, so the gain is not solely a routing-efficiency effect. All policies have zero deaths in that short horizon. Over 900 s, AR first death is 423.60 s battery-only versus 432.79 s with harvesting. For every harvested policy, the mean eighty_pct_death_observed flag is zero across five runs: none reaches the endpoint before the horizon. The generic saved-table phrase censored/mixed is therefore right-censored in all five harvested runs per policy. Battery-

only endpoints are observed. No sentinel or horizon value is used as an exact harvested death time, and no full Kaplan–Meier or hazard-ratio analysis is claimed.

E5/E5b: ablations and constrained learning

Full AR does not dominate its component removals. Without GAN, PDR is 85.63% versus full 83.90%; without CMDP it is 84.50%. No-energy-excuse endpoints match the full configuration here. Component removal changes route and observation distributions, so the effects are not independent contributions. No-detector TPR/FPR zeros and AUC 0.5 are defaults. A delivery target of 0.80 yields the highest descriptive floor-sweep PDR, while larger lambda at 0.90/0.95 does not ensure target satisfaction. Selecting the favorable target after looking at this sweep would require a new reserved evaluation cohort.

E6: protection and control accounting

AR has about 218.4 kB of control traffic and 30.9 kB of cryptographic overhead under the selected attack. Removing payload protection sets cryptographic overhead to zero and gives 84.03% PDR versus full 83.90%, with considerable variation. Differences in packet size can change radio events, routes, and observations, so delay differences are not isolated cryptographic CPU benchmarks. Payload protection is X25519/HKDF/AES-GCM; the experiment does not validate every routing control message as authenticated or establish formal anonymity.

Commissioning follow-up and historical consistency

The thirty-pair comparison gives AR +1.16 PDR points, CI [−2.97,+5.30], and 16/30 wins. This is a modest positive point estimate with an inconclusive interval. Its +32 ms delay difference also spans zero. Exploratory greedy and low-exploration candidates are retained below because excluding unsuccessful tuning would overstate the evidence. The earlier native five-run snapshot and earlier Python comparison differ in configuration and code state; neither is a clean causal before/after experiment. The Python +2.32-point gray-hole estimate motivates the investigation but does not override current native findings.

Evidence-to-claim audit

Candidate statement	Assessment and supported wording
AR is generally better than EAURP	Not established; report condition-specific estimates and intervals
The native simulation is more realistic than a single-probability model	It explicitly adds radio, packet, route, queue, energy, and observation mechanisms; field validity still needs calibration
Gray-hole detection is accurate	Not supported for this workload; native window TPR is low and AUC near chance
The LSTM improved routing	Not supported in the selected condition; persistence was selected in all five runs
Harvesting improves energy availability	Supported within the synthetic energy process; not a field battery-lifetime guarantee
Higher PDR floor guarantees delivery	Not supported; lambda reacts but measured PDR is nonmonotonic
All tests are complete	Incorrect; 863/905 runs and 179/181 conditions are included
Five independent master seeds were used	Incorrect; one review seed with five RngRun streams
All 180 contrasts prove a ranking	Incorrect; unadjusted exploratory contrasts, with repeated clean settings
Replay proves the protocol works in real time	Incorrect; replay visualizes a recorded run; JSON/CSV and repeatable experiments support quantitative claims

Limitations and prioritized next experiments

The main limitations are incomplete density replication, a small per-condition sample, unadjusted multiple comparisons, shared fleet learning, reused clean threshold-calibration data, continuing evaluation-time exploration, a synthetic harvesting process, and adapted rather than independently reproduced baseline implementations. No additional outcome is inferred from missing data or from visual appearance of a trace.

First complete the existing E2 plan without changing its original settings or silently replacing included checkpoints. Next reserve clean calibration and attack evaluation cohorts, compare simple size-conditioned thresholds and nonlearn-

ing routes, and measure whether sufficient forwarding evidence is available before optimizing neural capacity. For forecasting, retain persistence and test a longer, physically calibrated trace before attributing value to an LSTM. For routing, distinguish frozen evaluation from online adaptation and test a decentralized parameter-sharing alternative. Finally define a practical effect size, plan sample size and multiple-comparison control, and evaluate on independent settings. These are proposed experiments, not completed evidence.

Reproducibility, file inventory, and trace provenance

Artifact	Repository-relative location and purpose
Native routing and learning	ns3/contrib/ar-eaurp/model; actual C++ packet and model logic
Native unit-test source	ns3/contrib/ar-eaurp/test/ar-eaurp-test-suite.cc
Study settings	results/ns3-native-paper-review/study_settings.json
Condition settings	results/ns3-native-paper-review/configuration_settings.csv
Run settings and checkpoint hashes	results/ns3-native-paper-review/run_settings.csv
All E1–E6 aggregate outcomes	results/ns3-native-paper-review/csv
AR–EAURP paired estimates	results/ns3-native-paper-review/paired_e3_summary.json
All 180 E3 contrasts	results/ns3-native-paper-review/pairwise_e3_comparisons.csv
Commissioning follow-up and unsuccessful tuning	results/optimization_study; per-run JSON and logs
Earlier native comparison	results/final_review_ns3_results/FINAL_NS3_EVALUATION_REPORT.md
Earlier Python aggregate outcomes	output/full_results/csv
Presentation traces	results/ns3-live; inspect each run's XML together with its result.json

Trace XML is created by native ns-3 animation instrumentation while the original simulation executes. It records node positions and packet events. Uploading an XML file replays those recorded events with the parameters already used. Changing a live-run form does not change the saved trace. A replay control can make a demonstration easier without recreating a simulation, but results must remain attached to their original settings. The XML is useful for explanation; the saved JSON, raw checkpoints, hashes, and ledgers supply the numerical evidence.

To reproduce the main matrix from the project root in WSL:

```
bash ns3/scripts/run.sh --profile=review --only=e1,e2,e3,e4,e5,e6 --out=results/ns3-native-paper-review --render=0
```

To reproduce the separate commissioning comparison:

```
export NS3_HOME="$HOME/.cache/ar-eaurp/ns-3.45"
bash ns3/scripts/run_cmdp_commissioning_study.sh results/optimization_study/reproduction 21 50
```

These commands are instructions, not evidence of new execution during this revision. Exact checkpoint identity and source matching control resumability. A future completed matrix must state its new coverage and revision rather than relabeling the current partial report as complete.